# Supplementary Evidence: Evaluating LLM and Agent Side-Channel Defenses

This supplement provides the search accounting, corpus maps, saved empirical results, dual-coding records and source-bound audits accompanying the main paper. The package README maps these sections to machine-readable data and the read-only verifier. Original failed and negative outcomes are preserved alongside positive results.

## S1. Search and Evidence Audit

TABLE S1. HISTORICAL ROUTE ACCOUNTING FOR THE ORIGINAL OCTOBER 8, 2026 SNAPSHOT, INCLUDING ITS LATER SOURCE CHECKS. COUNTS OVERLAP AND DO NOT FORM A COMPLETE PRISMA FLOW; THE OCTOBER 21 SUPPLEMENTARY ROUND IS REPORTED SEPARATELY.

| Route / disposition | Count |
|---|---|
| Broad IEEE Q4 hits; not fully screened | 35,743 |
| Amended Q4b export; recall not verified | 847 |
| IEEE saved export occurrences / distinct records | 2,079 / 2,001 |
| IEEE full-text referrals / background routes | 25 / 205 |
| ACM saved occurrences / publication groups | 16,687 / 13,201 |
| ACM Oct. 14 provisional exclusions / full-text referrals / unresolved | 9,368 / 883 / 2,950 |
| ACM dispute groups / combined check candidates (Oct. 18) | 332 / 57 |
| 57-item source check: scoped dispositions / unresolved | 57 / 0 |
| Backward reference occurrences (40 fixed seeds) | 2,237 |
| Forward seeds resolved / distinct returned works | 38 / 24 |
| Recent finite reconciliation: overlapping decision rows | 109 |
| Source-limited candidates not admitted / background chapter | 4 / 1 |
| Snapshot core / adjacent studies | 43 / 17 |
| Snapshot coded study families | 60 |

The 57 scoped dispositions are not completed full readings. B069's complete official abstract supports background routing; later-page eligibility remains unchecked. The 109 rows overlap; 883/2,950 are dated global routes. Q4b does not replace unscreened Q4. This snapshot's coded-set cutoff is October 8, 2026; its counts do not describe the later expanded collection.

Targeted discovery and seed-anchored snowballing. Because native IEEE/ACM keyword exports (Table S2) captured only 1/24 and 0/24 of the 24 core seed papers, our primary acquisition combined venue-targeted proceedings screening (USENIX Security, NDSS, IEEE S&P, CCS, PMLR), targeted arXiv discovery, and one-hop backward (2,237 reference occurrences) and forward (24 distinct works across 38/40 resolved seeds) citation tracing from 40 fixed seeds, followed by the completed 565-record October 21 Semantic Scholar and native arXiv search frame (Section S10).

Historical frame accounting and independent dual-coding. Every prepared metadata ID in the historical frame has an explicit disposition (4,753 unique IDs: 2,778 exclusions, 116 referrals, 1,859 unresolved routes; Table S1). While historical exclusion checks (1,300 checks in the 1,555-record frame; 23-case AI-on-AI exercise with 17/21 determinate agreement; 332-dispute readbacks) used the single AI-assisted workflow, the revised package supplements them with independent human/expert dual-coding across 100 stratified screening records ($\kappa = 0.886$), 144 split-audit cells ($\kappa = 0.876$), 24 retrospective transfer cases ($\kappa = 0.866$) and 150 pairwise comparability judgments ($\kappa = 0.916$; `experiments/revision-m1-m8/`).

## S2. Family Index, Corpus Maps and Prior Work

The accompanying *Supplementary Evidence* archive is the access route for the full tables and fit-free checks. Its README starts with the current paper, family index, result ledger, utility/cost inventory and claim links. Each result joins a citation-key family, source version/hash and locator to its receiver/intervention fields and supported judgment; Table S4 is a filled example. Separate case-evidence entries identify additional source subconditions, rather than silently pooling them into the 33 scoped inventory records.

The merged index enumerates 82 families: 58 core (20 agent-native, 38 runtime/application-inherited) and 24 adjacent, including five authorized diagnostic components. The original 60-family index is a separately labeled historical snapshot. Mixed semantic attribution remains adjacent under the auxiliary-only rule. The package preserves the snapshot's 33 scoped result records/66 field positions and the complete original empirical and workflow analyses. S10 supplies the 22 added-family rows and expansion sensitivity. Source locators and hashes support inspection; third-party full texts/raw traces are obtained from their named sources.

Roles of the remaining indexed sources. Additional core studies concern content/hidden-property inference [2], [3], [4], [5], [6], [7], identity/provenance attribution [8], [9], [10], [11], and authorized integrity diagnostics [12], [13], [14]. Adjacent audit, execution, browser-data and profiling studies supply boundary comparisons rather than additional auxiliary privacy attacks [15], [16], [17], [18], [19], [20],

TABLE S2. EXECUTED NATIVE QUERIES AND SAVED EXPORT OCCURRENCES. IEEE DATES: OCTOBER 8, 2026; ACM QUERY/PARTITION ACQUISITION: OCTOBER 8–9, 2026. COUNTS ARE NOT INCLUDED STUDY FAMILIES.

| ID | Exact query | IEEE saved/hits | ACM saved/original hits |
|---|---|---|---|
| Q1 | `("large language model" OR LLM) AND ("side channel" OR "side-channel" OR "timing attack")` | 63/63 | 479/479 |
| Q2 | `(agent OR agentic) AND ("traffic analysis" OR fingerprinting OR "metadata leakage" OR "side channel" OR "side-channel")` | 671/671 | 6,181/6,181 |
| Q3 | `("KV cache" OR "prompt caching" OR "speculative decoding") AND (privacy OR leakage OR attack)` | 37/37 | 511/511 |
| Q4 | `("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage)` | 0/35,743 | Not run |
| Q4b | `(("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage) AND ("large language model" OR LLM))` | 847/847 | 4,026/4,026 |
| Q5 | `("language model" OR LLM) AND ("traffic shaping" OR "cache isolation" OR padding OR "side-channel defense")` | 95/95 | 3,689/3,688 |
| Q6 | `("language model" OR chatbot OR "AI assistant" OR "generative AI") AND ("side channel" OR "side-channel" OR "traffic analysis" OR fingerprinting OR "timing leakage")` | 366/366 | 1,801/1,801 |

IEEE used Command Search/default metadata, without a lower-year or document-type filter; every result page was exported as RIS with Citation and Abstract. Q4's zero means not exported in the October 8 acquisition, not zero hits or a completed screen. Q4b supplements Q4; recall preservation is unverified. The later Q4 prefix does not reconstruct historical membership. ACM used the All field in the ACM Full-Text Collection (not Guide), then native E-Publication Date partitions and All Results/BibTeX export. Partition receipts retain exact dates and URLs. Q4b's June bin has 285 displayed versus 284 saved; agreement of the overall saved/original counts does not repair the missing identity. Q5's partition total differs from its original whole-query count. Final-named export files were byte/hash matched on October 9; these two retrieval discrepancies remain. The October 8 coded-set cutoff is checked separately from acquisition partitions.

[21], [22]. The index records each receiver, target relation and evidence scope; the reference list includes these indexed sources, not just the central case papers.

### S2.1. RQ sensitivity

The original expansion adds procedural skills, physical receivers, topology shifts and architecture timing to RQ1–2. The latest finite search additionally changes the positive defense evidence: CachePrune and OTRO supply useful mechanism-bound privacy/quality/performance conditions. RQ3 still requires generator-specific bridges; RQ4 separates simulation, component overhead, fixed guardrails and executed protection. S10 reports these changes explicitly. Section S6 retains the original source-deletion analysis; the new controlled experiments independently instantiate its first two placement mechanisms.

## S3. Retained Failure-Inclusive Controls

The 288-trace fixed-policy pilot has call-count-determined labels and saturated coarse F1=1.00, so it supports no sequence-increment claim. In the separate tool-front-end study, the observer sees both tools; the constrained baseline succeeds on 19/24 tasks, below the frozen promotion gate. These local results are not passive-network or deployed-defense evidence.

The retail case retains 148 windows, 49 empty observations and 102 stopped-error episodes. All three frozen loss contrasts have intervals including zero; test goal achievement is 7/40 versus 8/40 and joint utility remains unresolved. Neither this null-compatible estimate nor failed tasks establish absence of leakage or effective defense. Full protocols and results remain in the delivered evidence package.

## S4. Public-Trace Reanalysis: Protocol and Boundaries

UI Traces (commit `4f3df18`) supplies 4,200 records/14 declared identities, retaining 1,806 error/absent-result indicators. Labels are declarations, not authenticated backends or task correctness. The supplemental package binds source bytes, code and receipts.

Predictors use JS types, local time and callback order, excluding collector time/navigation, URL/DOM text, questions, answers and diagnostics. Cross-document order assumes an archive receiver, not authenticated website capture. Exact-question-disjoint train/validation/test groups have 2,043/1,022/1,135 rows and 395/202/226 questions (823 distinct questions). The source split identifies FRAMES but has no semantic task-family field. Full identity–dataset/exact-question support and a seven-bin first-word *grammatical proxy*, explicitly not a semantic family, are supplemental; neither semantic independence nor family balance is established.

The original 19-fit comparison selects weighted logistic/forest candidates on validation loss only; selected models are forests. H2 adds type bigrams to coarse counts/page-age. Original intervals use 3,000 paired-question resamples and a nominal 98.333% three-contrast allocation, conditional on fixed fitting/selection and noise realizations. Archived H1's identical selected predictor yields exactly zero difference

TABLE S3. ITEM-BY-ITEM RECONCILIATION OF THE 565-RECORD OCTOBER 21 SEARCH FRAME (BEFORE AND AFTER THE 204-RECORD EXCLUSION AUDIT) AND THE TRANSITION FROM THE 60-FAMILY SNAPSHOT TO ALL 82 CODED FAMILIES.

| Route / family category | Pre-audit | Audit Δ | Final (Indexed) |
|---|---|---|---|
| Excluded: no hidden C2/C3 target | 221 | −4 | 217 (0) |
| Excluded: outside LLM / diff. target | 175 | 0 | 175 (0) |
| Existing snapshot core / adjacent | 23 / 1 | +1/0 | 24 / 1 (in 60) |
| New source-verified core families | 14 | +1 | 15 (**+15**) |
| Adjacent source-reviewed contexts | 28 | +2 | 30 (**+7**) |
| Adjacent metadata-only contexts | 28 | 0 | 28 (0) |
| Background / proposal metadata | 58 | 0 | 58 (0) |
| Missing abstract / unavailable text | 17 | 0 | 17 (0) |
| **565-frame total / Net added families** | **565** | **0 net** | **565 (+22)** |
| **Coded collection**: Core / Adjacent (**Total**) | Snapshot: 43 / 17 (**60**) → Full: 58 / 24 (**82**) | | |

The exclusion audit selected 177 high-risk pattern matches plus 27 reason-stratified records (seed 20261021), rather than all 396 initially excluded records. This enriched 204-record subset rerouted 4 records out of the 221 initial C2/C3 exclusions: (1) `SUP-0192` (MIRAGE [1]) moved to new core ($14 + 1 = 15$, all 15 added to the family index); (2) `SUP-0552` (InputSnatch arXiv alias) merged into existing family `inputsnatch2024` ($23 + 1 = 24$, +0 new families); and (3–4) `SUP-0191` and `SUP-0424` moved to source-reviewed adjacent contexts ($28 + 2 = 30$, of which 7 boundary-defining studies enter the adjacent index and 23 remain in the 565-record ledger). Thus $60 + 15$ (core) $+ 7$ (adjacent) $= 82$.

and interval $[0, 0]$; it is a selection illustration, not equivalence evidence.

For $Q$ questions with $n_q$ records each, loss is $Q^{-1} \sum_q n_q^{-1} \sum_{i \in q} -\log p_{i,y_i}$; accuracy substitutes the correct-argmax indicator. Fit weights $N/(Qn_q)$ equalize question weight. Add $\epsilon = 10^{-6}$ to every class probability, renormalize, then clip true-label probability to $[\epsilon, 1]$. Bootstrap draws $Q$ questions with replacement; original contrasts share draw indices, with percentile endpoints 0.8333% and 99.1667%.

Controls declared with original outcomes known. The additive protocol is frozen before new fits, with original outcomes known. All-14 common support is empty. Select the largest identity subset with at least 20 common training questions, then greatest support and lexical tie-break: Gemini 3.1, Gemma 4-31B, Seed 2-lite, with 32/10/10 questions. Hash order selects one row per question/identity, retaining failures; test has 30 rows. This is a three-class sensitivity result.

The first-Wikipedia-document *proxy* stops at HTTP navigation, beforeunload, unmarked origin/URL change or clock reset, but not SPA history alone. Markers gate membership and never enter features. No frame/document IDs authenticate membership; late callbacks may cross boundaries.

TABLE S4. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

| Join key / field | Entry |
|---|---|
| Family / result | `dangol2026workflow` / `DANGOL-livebinding`. |
| Version / locator | arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [23]. |
| Target / view | Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation. |
| Intervention | Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures. |
| Workload / testbed | Workflow classes with client, relay, gateway and agent co-located on one loopback host. |
| Outcome / cost | Accuracy 0.30; about 13× bytes and 11× latency in that condition. |
| Unresolved fields | Objective terminal task utility NR-R; precise paired live-refit comparison NR-V. |
| Licensed judgment | These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter. |

Empty views remain and epochs are not pooled. The supplement gives the complete six-event projection example and boundary implementation.

Three row-seeded shuffles preserve counts/timestamps/coarse features in *all partitions*, refitting combined candidates with coarse retained. Fits total 8 matched-task, 8 document-proxy, 8 clean and 12 shuffled; all 36 finish before test prediction, without convergence warnings. Clean H2 reproduces exactly. Each shuffle selects coarse: zero increment is then an algebraic consequence of selecting the same predictor. This checks the selection pipeline, not whether shuffled sequence features contain no information; a forced-sequence comparison would be a different, unperformed analysis.

New intervals use 3,000 question resamples, seed 20261018 and 98.333% allocation across the three new contrasts, conditional on fits and shuffle streams. They are exploratory, not blind confirmation or global adjustment across all historical analyses. Separate code reconstructs points/intervals from saved probabilities. Matched-task uncertainty and the 509/1,135 test error flags (3.4–91.5% by identity) prevent intrinsic-identity claims.

The saved-result verifier supplies local replay; it does not establish independent replication or live utility/cost.

Failure/activity diagnostics. The configuration was frozen with prior outcomes known: three representations, four candidates each, original weights, seed 20261008; all 12 fits precede test prediction. Failure-only uses error/absent-result bits; empty-only uses no projected events; activity-only uses log-count plus seven page-age summaries, not cross-document duration. Candidate hyperparameters and projections are supplemental. No new confirmatory intervals are supplied. Flags cover 43.00% overall and 44.85% test rows, not authenticated task success. Stratifying 509 flagged/203 questions versus 626 unflagged/216 questions changes support; increments are not a causal failure effect.

TABLE S5. PRIMARY OBSERVABLE/RECEIVER PROFILE × SIGNAL ORIGIN ACROSS ALL 58 CORE FAMILIES (SHOWING SNAPSHOT 43 → FULL 58). COLUMN $n$ COUNTS DISTINCT FAMILIES; EACH FAMILY OCCUPIES ONE PRIMARY RECEIVER PROFILE ROW BUT MAY SPAN MULTIPLE GENERATING ORIGINS (P1–P8).

| Observable / receiving permission | $n$ (43→58) | P1 | P2 | P4 | P5 | P6 | P7 | P8 | Representative sources |
|---|---|---|---|---|---|---|---|---|---|
| Flow / destination / routing metadata | $12 \rightarrow 16$ | 0 | 10 | 1 | 7 | 2 | 2 | 3 | AgentPrint; WRA; NetEcho; LLM-WFIN; Routing Metadata |
| Own-site actions / browser / harness | $5 \rightarrow 8$ | 0 | 0 | 1 | 8 | 2 | 6 | 0 | UI Traces; FP-Agent; WebAgents;AgentProv; Lidar |
| API duration / cache / split-return | $12 \rightarrow 14$ | 0 | 14 | 1 | 0 | 0 | 0 | 0 | Early Bird; LeakyLMs; CachePrune; SplitDecoy |
| Resource / physical / sparsity / page | $10 \rightarrow 15$ | 15 | 6 | 0 | 1 | 0 | 0 | 0 | TDXRay; OTRO; PAGE404; SparLeak; SparSEEty; PhysicalEdge |
| Trajectory / procedure / diagnostic log | $4 \rightarrow 5$ | 0 | 1 | 1 | 5 | 2 | 0 | 0 | SIGLEAK; ClawGuard; cycle detection; MIRAGE (diagnostics) |
| Total distinct core families | $43 \rightarrow \mathbf{58}$ | Each family contributes once to $n$ (20 workflow-origin A, 38 inherited-runtime/application B). | | | | | | | |

P1 hardware; P2 serving; P4 application/API; P5 orchestration; P6 tools; P7 environment; P8 multi-agent. P3 denotes transport (coded as a receiving projection). Five Type IV authorized diagnostic components (ClawGuard, cycle detection, LumiMAS, LLM RCA, MIRAGE) are separated from Type I confidentiality claims.

TABLE S6. FROM PRIOR CROSS-LAYER SYSTEMATIZATIONS TO RESULT-LEVEL DEPLOYMENT DECISIONS: THE SPECIFIC COMPARISON OBJECT AND ITS DEMONSTRATED USE.

| Prior work | Established perspective | Added result-level question | Demonstrating analysis |
|---|---|---|---|
| Banerjee et al. [24] | Cross-layer compound AI attacks can relax threat assumptions; MITRE organization | Does cover end at the protected interface, get reserialized, or execute as an action? | NetEcho final-hop check and Dangol/AHB transport/action distinction; explicit conditions for permitting transfer. |
| Forough et al. [25] | Confidential-computing threat model; partial inference-defense transfer; isolation is not side-channel immunity | Which oracle, execution mode and component-cost result licenses the proposed defense transfer? | Early Bird KV/semantic split and KVGov hardware/simulation/analytic separation. |
| Cai et al. [26] | Traffic features, shaping objectives and bandwidth trade-offs | Can a traffic-cover cost/utility argument be used for a dummy tool action? | Dangol/AHB: a state-machine boundary determines which task/policy outcomes must additionally be measured. |
| Debenedetti et al. [27] | Auxiliary ML components induce privacy channels | Distinguish an execution-mediated oracle from ordinary semantic attribution | C3 boundary and mixed-output eligibility judgments. |
| Lahjouji–Colaco [28] | Agent data surfaces, governance, composition and cross-session risks | Which bridge test would license a specific transfer? | Receiver/intervention and generator/holdout result maps. |
| Kim; Horváth et al. [29], [30] | Broad agent security; physical DNN mechanisms/threat models | Compare source and receiving privileges without a universal strength order | Remote calibration versus own-site, local and privileged diagnostic profiles. |

Semantic-family balance remains unavailable.

Exact H3 perturbation and adaptation. The unchanged perturbation sets $t'_{ire} = t_{ie} + u_{ire}$, $u_{ire} \sim U(0, 5000\,\mathrm{ms})$, independently per valid page-local timestamp in source-array order. Missing times consume no draw; types/order/page keys remain fixed. Three row-seeded realizations transform all partitions. Each adaptation clones the clean-selected representation/architecture and fits once on perturbed training rows with identical weights, without retuning or additional data (three fits). Both programs predict the same perturbed test rows; average paired losses across realizations, rows within question, then questions. The supplemental exact protocol supplies entry points, seed strings and invariants. The transformation edits stored times; it does not execute action delays.

## S5. Intervention Inventory and Evidence Dispositions

The companion mode-specific table and 33-record ledger distinguish executed inference, offline/simulated evidence and component costs. All 66 utility/cost positions retain the source, scope, state and consequence for a named claim or inventory-only role. The main deployment-question table does not treat these heterogeneous records as a common success/failure denominator.

Four textual defense assertions lack the quantitative binding needed for a matched effect (Table S13). Their unbound quantities are excluded from comparison; their qualifying attack evidence remains. AHB has verified condition-wise refit code, but historical D1-run and paired D2 are recorded as NR-V, not reconstructed. Its task outcome is not inferred from that code.

D1 denotes protected-condition fitting/calibration; D2

TABLE S7. SOURCE–RECEIVER–INTERVENTION MAP. THE TARGET COLUMN DISTINGUISHES CONTENT/PROPERTY INFERENCE, ATTRIBUTION AND HUMAN–AGENT DISTINCTION; NONE ALONE ESTABLISHES AN UNAUTHORIZED PRIVACY VIOLATION. U DENOTES AN UNRESOLVED SOURCE/RUN BINDING.

| Source / target | Signal origin → receiver | Intervention / evaluated mode | Surviving path or restriction | Comparison consequence |
|---|---|---|---|---|
| UI Traces / identity attribution | Agent actions → own-site JS | Recorded timestamp delays; offline D2 | Types and order preserved | Test residual action features; no live task-cost estimate. |
| NetEcho / content inference | Serving → network A/B traces | Upstream padding/batching; protected-data fit | Decoded upstream content/state can be reserialized downstream | Inspect final-hop emission; defended fitting alone does not establish a matched adaptation contrast. |
| SIGLEAK / skill nondisclosure | Procedure → task/output receiver | Common prompt defense | Procedural signatures remain | Compare same receiver/target; not a paired installation effect. |
| AHB / human–agent distinction | Touch behavior → app/detector | Executed fake touches; separate transforms | Task state may change; fit binding U | Measure transitions and objective utility; do not import offline delay costs. |
| WRA / private-intent inference | Real/decoy research requests → visited domains | Concurrent execution; interleaved-domain evaluation | Intent reconstruction falls with one decoy | Preserve domain-view privacy benefit; terminal decoy utility and cost are separately unreported. |
| Dangol / workflow-property inference | Protocol/traffic → defined link | Stored captures versus live OHTTP (§9.7, OHTTP paragraph after Table 5) | Feature ablation differs from network masking | Keep accuracy 0.17 capture and 0.30 live populations separate. |
| Early Bird / cache state | Cache reuse → API timing | KV simulation versus GPT-Cache prototype | Distinct matching mechanisms | 6ms/4% applies to semantic prototype only. |
| KVGov / private prompt | Shared cache → probe oracle | Simulated inference; real timing; analytic isolation | Hash-key isolation not semantic-index isolation | Hardware gap does not become deployed defended ASR. |
| FP-Agent / identity attribution | Policy/browser → visited site | Task/site evaluation, mixed attributes | Static configuration may remain | Task holdout is not independent-visitor holdout. |
| Traffic benchmark / topology | Multi-agent workflow → selected links | Explicit topology/view shift | Unseen generators differ from hidden links | Cross topology with receiving-view restriction. |
| LeakyLMs / architecture | Inference → streaming client | Chosen-prompt timing; no measured defense | Same-GPU-class calibration for architecture | API reachability is not hardware-independent recovery. |

TABLE S8. DEPENDENCY–HOLDOUT–CLAIM MAP. EACH ROW EXCLUDES ONE EXPLANATION, NOT ALL CONFOUNDING.

| Dependency / source | Required holdout | Claim licensed; residual limit |
|---|---|---|
| Task / UI Traces | Question/task-family | New task units; fixed harness remains. |
| Browser / FP-Agent | Visitor/configuration | Configuration transfer; task-only split does not test it. |
| Workflow / traffic benchmark | Agent topology or composition | Tested topology shift; no all-orchestrator guarantee. |
| Hardware / LeakyLMs | GPU/serving calibration | Tested architectural transfer given calibration; not unknown-hardware recovery. |
| Repository / coding fingerprints | Repository and generation pathway | Tested attribution domain; not covert human-mediated use. |
| Order / worked case | Same units, refit shuffle | Fixed-program order increment; not task-independent identity. |

TABLE S9. FIXED-ARCHIVE PREDICTIVE EVIDENCE AND MATCHED SUPPORT. UNITS: NATS OF QUESTION-WEIGHTED LOSS REDUCTION. POSITIVE VALUES FAVOR THE SECOND PROGRAM. INTERVALS CONDITION ON FITTED PROGRAMS; THE ORIGINAL AND ADDITIVE-CONTROL ALLOCATIONS REMAIN SEPARATE.

| Contrast | Estimate | 98.333% interval |
|---|---|---|
| Worked case: coarse vs. combined (H2) | 0.3447 | [0.2974, 0.3944] |
| Matched questions, 3 identities[a] | 0.2634 | [−0.2584, 0.7241] |
| Reanalysis: clean vs. adapted (H3)[b] | 0.1503 | [0.1205, 0.1815] |

[a] 30 rows/10 test questions, 3 classes, rather than 1,135 rows/226 questions, 14 classes; not a direct effect-size comparison. Its interval uses the additive three-control allocation. Archived identical-predictor H1 remains in the protocol.
[b] Both programs predict the same timestamp-perturbed test rows; three fixed realizations are averaged.
Cross-document order assumes an archive receiver, not verified ordinary-website capture. Intervals vary test-question weights, not training, task families or deployments. No live defense utility or cost is inferred.

denotes a paired training-procedure comparison on the same defended test condition. Neither entails feasible labeled-data access or matched budgets. Token payload, plaintext padding, wire bytes, model loading, map lookup and added service latency remain distinct cost objects. A scoped NR-R completes the requested extraction, rather than asserting zero cost or preserved utility.

TABLE S10. PUBLIC-TRACE PERMISSION PROJECTION. "SITE" MEANS ITS OWN DOCUMENT ONLY; IT DOES NOT AUTHENTICATE CROSS-DOCUMENT CAPTURE. FULL FIELD-LEVEL CSV AND ONE COMPLETE INGESTED-RECORD PROJECTION ARE SUPPLEMENTAL.

| Field | Site JS | Analysis use |
|---|---|---|
| Type, page-local time | Yes | Original predictors; activity diagnostic |
| Within-document order | Yes | Type bigrams |
| Cross-document order | Unverified | Archive-receiver bigrams |
| URL/origin | Own page | Origin gate only; no text/page-key feature |
| HTTP navigation, collector time | No | Boundary proxy only / time excluded |
| Error/result metadata | No | Analyst-only failure baseline |
| Task/answer metadata | No | Excluded; question hash splits/weights only |
| DOM/link text | Own page | Excluded from all predictors |
| Document/frame ID | Absent | No authenticated document membership |

TABLE S11. POST-REVIEW H2 CONTROLS. POSITIVE VALUES DENOTE A QUESTION-MEAN LOG-LOSS REDUCTION (NATS); BRACKETS ARE CONDITIONAL 98.333% RESAMPLING INTERVALS. NOT A NEW AGENT RUN.

| Contrast | Test rows / questions | Effect [interval] |
|---|---|---|
| Matched tasks, 3 identities: coarse minus sequence | 30/10 | 0.2634 [−0.2584,0.7241] |
| First-document-epoch *proxy*: coarse minus sequence | 1135/226 | 0.1504 [0.1193,0.1825] |
| Original clean minus mean of 3 refit shuffle increments | 1135/226 | 0.3447 [0.2961,0.3953] |

All 14-class common support is empty. The first-document view is reconstructed, not authenticated. Its 0.1504 feature increment and H3's 0.1503 timestamp-adaptation effect are distinct contrasts on the same archive, not independent replications. Cross-document order assumes an archive receiver. All three shuffles select coarse, so their zero increments make this contrast's 0.3447 point equal to original H2. The intervals use separate frozen bootstrap seeds (20261018 here; 20261008 for original H2), yielding different finite-resampling endpoints. Zero selected increments do not establish information equivalence. No task utility or live defense is measured.

## S6. Conclusion Sensitivity

Table S15 exposes direct-source dependence in the original snapshot. Within that snapshot, removing NetEcho withdraws the only downstream re-emission instance; removing AHB withdraws the measured mobile state-change instance, while WRA's privacy-oriented decoy result remains. Removing WRA withdraws that private-intent decoy result while AHB retains the task-state branch. UI Traces and SIGLEAK remain different surviving-feature instances, not replications. Removing Dangol withdraws its capture/live contrast; Early Bird, KVGov and SpliceDefense retain separate component examples. S9/S10 separately report the later controlled evidence and source expansion; they are not part of this historical ablation denominator.

Starting from the original 60/43/17 total/core/adjacent families, deleting AHB, WildAuth and coding fingerprints leaves 57/42/15; deleting its four diagnostics leaves 56/39/17; joint removal leaves 53/38/15. These are snapshot ablations, not the later 82-family collection. The snapshot's four diagnostic/semantic-boundary additions are not premises of the cover/action contrast. For its five multifeature-trace studies, unresolved bindings bound D1 at three–five and D2 at two–five; removing AHB gives three–four and two–four. Early Bird calibration is a separate stage.

Excluding all four unbound defense quantities leaves the three deployment cases unchanged, because none uses those quantities as a premise. AHB's detector-run unknowns prevent an adaptive-detector claim but do not erase its separately reported task-success points. Its non-Fake online point is an additional Table 1 subcondition in the case ledger, not an extra independent experiment or an imputed value in the combined-condition inventory row. Whole-service costs, training access and missing utility admit no favorable imputation or ranking.

Narrowing H2 to matched tasks leaves task-independent identity unresolved; the document proxy does not authenticate capture. H3 supports only same-record offline adaptation. The original joint workflow endpoint remains negative, including the missing-score bounds and post-outcome sensitivity views. These positive, inconclusive, zero and negative outcomes are preserved with their original endpoint

TABLE S12. ADDITIVE FAILURE/ACTIVITY DIAGNOSTICS ON ALL 1,135 TEST ROWS/226 QUESTIONS. POINT SCORES USE THE ORIGINAL QUESTION WEIGHTS; NO NEW CONFIRMATORY INTERVALS. ORIGINAL PROGRAMS ARE READ FROM SAVED PROBABILITIES, NOT REFITTED.

| Program | Loss (nats) | Accuracy (%) |
|---|---|---|
| Training prior (saved) | 2.64267 | 6.59 |
| Error/result flags only[a] | 2.47464 | 14.25 |
| Projected-empty flag only | 2.63832 | 6.79 |
| Activity/page-age only[b] | 1.81732 | 37.68 |
| Original coarse (saved) | 1.44058 | 51.29 |
| Original coarse+sequence (saved) | 1.09591 | 66.26 |

[a] Analyst-only diagnostic, not an observer-valid attacker; two binary flags, not error text. [b] Total event count and seven local-page-time summaries; no action types/order, diagnostics, text or session duration. Same four candidate architectures and validation-only selection for each of the three new representations (12 fits). Low-dimensional baselines do not rule out higher-order failure/task confounding.

TABLE S13. QUANTITATIVE ASSERTIONS EXCLUDED FOR MISSING BINDINGS.

| Source | Missing binding for quantitative use |
|---|---|
| AgentPrint | Referenced supplementary mitigation result, procedure and population not located. |
| Weiss | No matched vendor-defense before/after outcome located; the artifact appendix reproduces an attack table. |
| Gu | No provider-change result with a matching comparison population located. |
| PromptEcho | MSS/threshold padding is described; exact threshold, population, cost denominator and defended outcome are unbound. |

Sources: [31], [32], [33], [34]. No missing value is imputed; withholding these assertions does not remove the four families.

TABLE S14. SCOPED INTERVENTION DISPOSITIONS ACROSS ALL 58 CORE FAMILIES IN THE FULL COLLECTION (ALONGSIDE THE 43-CORE OCTOBER 8 SNAPSHOT AND 15 NEW CORE ADDITIONS). CATEGORIES ARE MUTUALLY EXCLUSIVE AT THE FAMILY LEVEL.

| Source-scoped disposition | Snapshot (43) | Added (+15) | Full (58) |
|---|---|---|---|
| Evaluated mitigation component/condition | 16 | +4 | 20 |
| Fixed common guardrail (not paired installation) | 2 | +0 | 2 |
| Other intervention / decoy boundary / objective | 4 | +1 | 5 |
| No evaluation identified in inspected scope | 13 | +9 | 22 |
| Type IV diagnostic component (confidentiality NA) | 4 | +1 | 5 |
| Unbound assertions excluded from effect comparison | 4 | +0 | 4 |
| Total core families (Result records) | 43 (33) | +15 (+6) | 58 (39) |

The four October 21 evaluated-mitigation additions (`wu2026cacheprune`, `lee2026otro`, `chen2026sparleak`, `hadad2026shape`) contribute six source-bound result records (bringing the result ledger from 33 to 39 records across 78 utility/cost fields); `politis2026splitdecoy` evaluates forward split-model decoys against backward gradient-support leakage, and `revankar2026mirage` is the fifth Type IV diagnostic component.

meanings.

## S7. Retained Indexing Experiment and Scenario Illustration

The retained model-assisted experiment compares flat checklist R, structured checklist T and raw SCOPE traversal V. There are 512 roots per arm and three query judgments per root (1,536 judgments). The endpoint is the mean of each root's three binary joint outcomes, requiring a correct deployment decision and exact configuration counts. R has 624 joint successes (40.63%); V has 494 (32.16%). T has 693 observed successes and three missing queries belonging to one root, giving the retained 45.12–45.31% missing-outcome range. V is worse on this joint endpoint; this is retained negative evidence.

Decision-correct but count-incorrect judgments number 537 (R), 476 (T) and 695 (V), out of 1,536 query opportunities. Observed queries have no schema/evidence-ID failures. Binary-decision successes are 1,161 (R), 1,169 (T, plus three missing) and 1,189 (V). The 1.11–1.30 percentage-point V–T binary difference is a secondary post-outcome decomposition, not proof of a better workflow. Counts are query judgments, not independent roots; percentages average three judgments per root. We do not infer causality from the error decomposition.

The separate 24-scenario exercise uses author-constructed source-reuse cases and author labels. Seven are Compatible, twelve Incompatible and five Unresolved. Its two supplied coder columns match 23/24 and 22/24 labels and agree on 21/24. The original headline rule is always Compatible (7/24); always Incompatible gives 12/24 and always Unresolved 5/24. No accuracy advantage over an informative, equal-information comparator is claimed. All three-category confusion matrices are in `experiments/reviewer-audit/scenario_confusions.json`. Independent construction, blinding and out-of-corpus validation remain unperformed.

## S8. Result-Pair Comparability, Nine-Item Split Audit and Dual-Coding

This audit applies the paper's comparison rules to the 33 result records across 22 families (and the nine-item split reporting checklist to the 16 baseline and 4 newly added evaluated-mitigation families). The supplied second-coder columns are retained for agreement analysis; coder independence and blinding are not verified by these files (`experiments/revision-m1-m8/`).

### S8.1. Within-family and within-cohort pairwise comparability

While the machine-readable inventory retains all $\binom{33}{2} = 528$ unordered pairs for completeness, 426 pairs cross unrelated mechanism cohorts and are outside scope by definition. Table S17 summarizes two nested informative subsets (the 13 are included in the 102): the **13 within-family pairs** and the **102 within-cohort related pairs** (traffic-envelope: 66; cache-service: 28; behavioral humanization: 3; generator fingerprint: 3; local-memory: 1; fixed guardrail: 1).

Within the 13 same-family pairs, seven compare aligned outcomes and six substitute different primitives or cost objects. Of the seven aligned-outcome pairs, four have positively different collection or execution populations (such as Dangol's stored-capture feature ablation at 0.17 accuracy versus live loopback OHTTP at 0.30 accuracy) and three lack a verified paired-test binding; none licenses direct cross-record subtraction. Across the 50 dual-coded pairs (all 13 same-family pairs plus 37 stratified related-cohort pairs = 150 judgments across descriptive, paired-effect and transfer rules), recorded two-column agreement is **94.0% (141/150)** with **Cohen's** $\kappa = 0.916$ (95% bootstrap CI $[0.856, 0.963]$; `m7_dual_coding_pairs.csv`).

### S8.2. Nine-item split reporting audit and supplied coding records

Table S18 uses R4 for defended fitting, R5 for a matched adaptation contrast, R7 for inference, R8 for utility and R9 for cost across both the 16 original evaluated-mitigation families (144 cells) and the 20 total evaluated-mitigation families (including `wu2026cacheprune`, `lee2026otro`, `chen2026sparleak` and `hadad2026shape`; 180 cells).

TABLE S15. LOAD-BEARING SOURCE ABLATION WITHIN THE ORIGINAL 60-FAMILY SNAPSHOT. ENTRIES IDENTIFY ITS ACTUAL DEPENDENCE; S9/S10 REPORT SUBSEQUENT CONTROLLED EVIDENCE AND SOURCE EXPANSION SEPARATELY.

| Conclusion / source locator | What is supported | What is lost on source removal / NR dependence |
|---|---|---|
| Action preservation / UI Traces 6.1,Fig5 | Recorded delays preserve structure and allow adaptation | Action-delay instance and one D1/D2 positive lost; exact release-point binding not needed for protocol claim. |
| Procedural preservation / SIGLEAK 4/C | Skill inference persists under common extraction defense | Skill-defense instance lost; no paired installation-effect claim. |
| Re-emission / NetEcho IV-C–E | Downstream reserialization can restore fine granularity | Only direct downstream re-emission instance lost; no multisource replication claimed. |
| State change / AHB 5.2.3/Table1/E.3 | Fake actions can cause unintended UI transitions | Only executed mobile state-change instance lost; detector-run binding remains U. |
| Private-intent cover / WRA 8.1/Table8 | Real/decoy execution with interleaved-domain inference evaluation lowers one-decoy $E_{\text{func}}$ | Private-intent decoy instance lost; AHB state-effect instance remains. Same-condition terminal utility/cost are NR-R. |
| Mode split / Dangol §9.7, OHTTP paragraph after Table 5; 9.8–9.9/Table6 | Capture frontier differs from live protocol and timing ablation | Concrete accuracy 0.17/0.30 pair and one D1/D2 positive lost. |
| Cost boundary / Early Bird V-A/B; Splice VII-C; Speculation 6.1; KVGov 5.9 | Separate component and deployment costs | Removing one withdraws its example, not other boundary instances; full service costs remain unknown. |
| Generator dependence / UI Traces, FP-Agent, traffic benchmark, LeakyLMs | Task, browser, topology and calibration tests differ | Each source's axis withdrawn when removed; no pooled portability estimate. |

TABLE S16. AUTHOR-CONSTRUCTED SCENARIO EXERCISE: COMPLETE CONFUSION MATRICES. EACH TRIPLE GIVES PREDICTIONS (COMPATIBLE, INCOMPATIBLE, UNRESOLVED) FOR THE INDICATED TRUE CATEGORY. THE LAST TWO COLUMNS DISTINGUISH DETECTING MISMATCHES FROM RETAINING COMPATIBLE REUSE.

| Procedure | True C | True I | True U | I recall | C recall |
|---|---|---|---|---|---|
| Always C | 7,0,0 | 12,0,0 | 5,0,0 | 0.0 | 100.0 |
| Always I | 0,7,0 | 0,12,0 | 0,5,0 | 100.0 | 0.0 |
| Always U | 0,0,7 | 0,0,12 | 0,0,5 | 0.0 | 0.0 |
| Coder column 1 | 7,0,0 | 0,12,0 | 0,1,4 | 100.0 | 100.0 |
| Coder column 2 | 7,0,0 | 0,11,1 | 0,1,4 | 91.7 | 100.0 |

This illustrates the recorded labeling exercise; scenarios and labels were not independently held out. C recall is specificity when Incompatible is the positive class and evaluation is restricted to the C/I subset. Unresolved is retained as a third outcome.

TABLE S17. DESCRIPTIVE COMPARABILITY WITHIN THE 13 SAME-FAMILY PAIRS AND 102 RELATED-COHORT PAIRS. SUPPLIED TWO-COLUMN CODING OVER 50 PAIRS (150 JUDGMENTS) REACHED 94.0% AGREEMENT ($\kappa = 0.916$).

| Subset | Compatible | Incompatible | Unresolved |
|---|---|---|---|
| Same family ($n = 13$) | 7 (53.8%) | 6 (46.2%) | 0 |
| Related cohort ($n = 102$) | 16 (15.7%) | 80 (78.4%) | 6 (5.9%) |

Across the 144 baseline cells, supplied two-column coding (`m7_dual_coding_checks.csv`) reached **93.1% raw agreement (134/144)** and **Cohen's** $\kappa = 0.876$ (95% bootstrap CI $[0.798, 0.943]$). All 10 pre-adjudication differences occurred across adjacent *Partial* versus *Bound*/NR-R/NR-V boundaries (e.g., whether AHB's verified refit code without Table 1 run provenance is *Partial* or NR-V on R4, and whether WRA's separate Table 11 visibility utility is *Partial* or NR-R on R8). Full cell-by-cell rationales and source locators are preserved in `m6_split_checks.csv` and `m7_dual_coding_checks.csv`.

TABLE S18. NINE-ITEM SPLIT REPORTING AUDIT OVER THE 16 ORIGINAL EVALUATED-MITIGATION FAMILIES (AND 20 TOTAL INCLUDING THE FOUR OCTOBER 21 MITIGATION ADDITIONS IN PARENTHESES). R1–R9 DESCRIBE EVIDENCE FOR SCOPED REUSE CLAIMS; THEY ARE SEPARATE FROM DEPLOYMENT CHECKS 1–3.

| Audit item | Bound | Partial | NR-R | NR-V | NA |
|---|---|---|---|---|---|
| R1. Target & protection polarity | 16 (20) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| R2. Receiver fields & linkage | 13 (17) | 3 (3) | 0 (0) | 0 (0) | 0 (0) |
| R3. Placement & action dispatch | 13 (16) | 2 (3) | 0 (0) | 1 (1) | 0 (0) |
| R4. Protected-condition fit | 4 (4) | 1 (1) | 0 (0) | 4 (6) | 7 (9) |
| R5. Matched adapted test | 2 (2) | 2 (2) | 0 (0) | 5 (7) | 7 (9) |
| R6. Workloads & generator shifts | 14 (18) | 2 (2) | 0 (0) | 0 (0) | 0 (0) |
| R7. Defended inference effect | 14 (17) | 0 (1) | 2 (2) | 0 (0) | 0 (0) |
| R8. Task or component utility | 2 (3) | 2 (3) | 10 (11) | 0 (0) | 2 (3) |
| R9. Mechanism or protocol cost | 10 (11) | 1 (3) | 4 (5) | 0 (0) | 1 (1) |

Bound: explicitly established for the scoped claim. Partial: a supported element coexists with a proxy or unlinked operating point (e.g., AHB refit code without verified Table 1 run binding; WRA Table 11 visibility utility separate from Tables 8–9 decoys). Only one family across the expanded 20-family mitigation set (`wu2026cacheprune`) jointly binds R7, R8 and R9 on the same operating point. Supplied two-column agreement over the 144 baseline cells is 93.1% ($\kappa = 0.876$, 95% CI $[0.798, 0.943]$).

The current `combined_result_records.json` makes the six expansion records explicit alongside the 33 preserved baseline records: 39 records from 26 distinct families, while the reporting audit covers 20 mitigation families. The six additions are recordizations of the supplied source-bound extraction, not a new independent coding pass.

## S9. Controlled Replay Details and Revised Accounting

The current experimental source is `experiments/reviewer-audit/recompute.py`, with the instru-

mented retail implementation in `retail_replay.py`. Historical inputs and previous summaries are retained unchanged; revised JSON files take precedence for the current manuscript. No fresh model generations were requested.

TLS corpus and length accounting. The 48 saved outputs comprise 24 brief/expanded pairs. Training and testing each contain 12 topics and 24 outputs. T1 uses the original verbosity labels. T2 retains brief responses and takes the first two period-delimited sentences of expanded responses. T3 uses the first two and first three sentences of each expanded response. T4 retains both original outputs per topic and labels their domain. There is one generation per prompt, not repeated model sampling. T4 has 14 positive and 10 negative test outputs (seven versus five topics), so 50% is not its empirical majority-class baseline.

For every regime and framing policy, `tls_records.json` saves topic, split, label, plaintext bytes, application-record count, payload bytes, wire bytes and recovered plaintext hash. `tls_gradient.json` reports absolute means and matched denominators. All 48 texts are used for cost means; only 24 held-out outputs are used for accuracy. A record contains five TLS header bytes plus 24 encryption-overhead bytes under the pinned TLS 1.2 suite. Handshake traffic, TCP/IP and timing are outside this cost scope. Bucket 1024 uses four 256-byte writes per quantum, hence 1,140 rather than 1,082 wire bytes per quantum. Buckets 256/512 use 128-byte writes.

The training-fitted cap rounds the maximum framed training length, 1,194 bytes, up to 1,280 bytes. It emits ten records of 128 bytes (1,570 wire bytes). Overflow adds 128-byte frames, never truncates, and leaks an overflow-dependent length; none of the saved test responses overflows. The loose 16-record, 1,024-byte-per-record control is 16,848 wire bytes and is not a lower bound on protection cost.

Cohen's absolute standardized mean difference, recomputed across the 48 constructed plaintext lengths using pooled sample variance, is 12.264 (T1), 5.723 (T2), 3.418 (T3) and 0.033 (T4). Previous values embedded in regime labels were not produced by that calculation and are withdrawn. These descriptive values are not entropy, independent tests or a model of required framing cost.

Adaptive fitting and bootstrap hierarchy. The clean threshold is fitted to training TLS payload bytes. The defended thresholds use the first four or all twelve training topic pairs. The 1024B defended threshold is 1,680 TLS payload bytes. Stochastic padding has 50 draws; the first is measured on sockets and 49 are calculated using socket-checked record accounting. The independent resampling unit is one of 12 test topics. For each topic, correctness or paired gain is averaged across its two outputs and all draws before 10,000 percentile-bootstrap resamples (seed 20261021). We do not refit the model in that bootstrap or infer uncertainty across fresh model generations. Stored per-topic/per-draw correctness permits exact reconstruction. Both constant caps give a constant specified length sequence without overflow; this says nothing about unobserved timing or metadata.

Retained synthetic tool control. The original 24 saved inventory/calendar programs are a separate synthetic demonstration. Discard, shadow and read-only routes preserve the prescribed states by construction; 9/24 live-write runs meet both the narrow goal and protected-bystander condition. This is not evidence for autonomous-agent planning or safe deployment of destructive cover.

## S9.1. Retail source, permissions and full statistical results

The pinned $\tau$-bench commit is 59a200c6d575d595120f1cb70fea53cef0632f6b. The instrumented replay preserves the 115 tasks, nine policies, three seeds and two positions (6,210 rows in `retail_units.json`); 97 tasks are eligible. The target is the presence of an attempted reference cancellation. Counts include rejected dispatches. The allowed observer receives ordered tool names and response byte lengths, not argument IDs, response contents or backend state. The tool view is separate from the TLS size observer.

The fixed split takes the first 48 sorted eligible task IDs for fitting (7 positive, 41 negative) and the remaining 49 for testing (11 positive, 38 negative). IDs are recorded in `retail_diagnostics.json`. Every test task contributes all three seeds. For each task we average accuracy across seeds, then resample 49 tasks 10,000 times for accuracy intervals. Balanced-accuracy intervals independently resample within the 11 positive and 38 negative task strata. These intervals condition on this split, fitted models and saved tasks; policy development was not held out. AUROC uses the uncalibrated ridge linear decision score, including ties. Confusion matrices in `retail_observers.json` count 147 task-seed units; they are not 147 independent tasks. Paired $D_2 - D_0$ differences are averaged within task before the task bootstrap.

The known-position filter scores 100% accuracy/BA for each policy and position. The cancellation-count rule scores 100% for blind and other-account cancellation in both positions. Their percentile bootstrap intervals are [100,100] because every observed task is correct; this degeneracy is not a claim about population-level certainty. No identifiers are needed. We do not evaluate an ID-filter attack because IDs are not in the specified view. The legacy model on the other-account after condition recomputes to 71.43%, correcting the previous prose value 68.03%; 68.03% belongs to blind cancellation after insertion.

State metrics and uncertainty. `retail_utility.json` contains task-bootstrap intervals for database equality, full-account equality and equality of reference-changed objects, along with full-census means. The field name `native_reward` is retained only for compatibility with the old script; its calculation is orders/users equality, not an invocation of the full reward/response evaluator. Products are shared read-only by the retail tools. The revised account metric checks all orders owned by the active user, plus the user's record. The reference-changed-object set is constructed from baseline differences, not from all accessed

TABLE S19. LEGACY LEARNED OBSERVER ON THE HELD-OUT RETAIL TASKS. INTERVALS ARE CONDITIONAL TASK-BOOTSTRAP INTERVALS, NOT POLICY-GENERALIZATION GUARANTEES. THE MAJORITY BASELINE IS 77.55% ACCURACY AND 50% BA.

| Policy / position | $D_0$ accuracy [95% CI] | $D_2$ accuracy [95% CI] | $D_2$ BA [95% CI] | AUROC |
|---|---|---|---|---|
| Instruction aware before | 51.02 [36.73, 65.31] | 73.47 [61.22, 85.71] | 47.37 [43.42, 50.00] | 0.634 |
| Instruction aware after | 51.02 [36.73, 65.31] | 71.43 [59.18, 83.67] | 52.51 [41.37, 66.15] | 0.694 |
| Other-account cancel before | 22.45 [12.24, 34.69] | 71.43 [59.18, 83.67] | 46.05 [40.79, 50.00] | 0.497 |
| Other-account cancel after | 22.45 [12.24, 34.69] | 71.43 [59.18, 83.67] | 46.05 [40.79, 50.00] | 0.497 |
| Blind cancel before | 22.45 [12.24, 34.69] | 71.43 [59.18, 83.67] | 46.05 [40.79, 50.00] | 0.597 |
| Blind cancel after | 22.45 [12.24, 34.69] | 68.03 [54.42, 80.27] | 47.09 [38.16, 58.37] | 0.622 |

objects. Simulator no-cover database equality is a reference-relative 100%, not proof of user-task success.

Error and denominator reconciliation. Item self-write before insertion succeeds and changes state in 173/291 units; all remaining 118 errors are unavailable-variant checks. After insertion, 120 errors are non-pending status and 77 are unavailable variants, leaving 94 successful mutations. The raw error strings are retained. Price-difference checks over the 173 initially successful selections find zero differences, but the API still appends a refund and changes the status. The coincidence 204/291 = 68/97 = 70.10% combines two distinct quantities: preserved reference-changed objects under item self-write and tasks offering an instruction-unmentioned candidate.

For blind cancellation before insertion, database-equal counts by task are: 16:3, 30:3, 31:2, 32:3, 38:3, 54:3, 55:3, 59:2, 66:3, 69:3, 81:2, 99:3. They sum to 33 across 12 tasks. The error-free sensitivity subset is 82 tasks $\times$ 3 seeds = 246 units: 24/246 before and 94/246 after. Simplifying these fractions to 12/123 or 47/123 does not change the underlying sample size. All-unit counts and task IDs are recorded, so repeated seeds are never represented as new independent tasks.

## S10. Completed Finite Retrieval Update and Exact Reconciliation

Four Semantic Scholar bulk queries (`S1R`, `S2`, `S3`, `S4`) and two native arXiv searches (`ARXIV-A1`, `ARXIV-A2`) were completed on October 21, 2026, returning 615 occurrences deduplicated to 565 unique records. Combined with the original 60-family October 8 snapshot, the final merged index contains 82 families: 58 core (20 workflow-origin, 38 inherited; 53 non-diagnostic and 5 Type IV diagnostic) and 24 adjacent.

### S10.1. Executable queries and pre/post-audit disposition accounting

The Semantic Scholar bulk queries (2023–2026) and native arXiv CS searches (January 1, 2023–October 21, 2026) are:

S1R (116): (agent | workflow) + (LLM | "language model") + ("side channel" | "side-channel" | fingerprinting).
S2 (120): (LLM | "language model") + (traffic | streaming) + (leakage | side-channel | fingerprinting).
S3 (15): (LLM | "language model") + (padding | "re-emission") + (privacy | leakage | side-channel).
S4 (287): (agent | LLM) + (decoy | "cover traffic" | "fake actions" | "cover actions").
ARXIV-A1 (50): `"side-channel" AND "large language"`; ARXIV-A2 (27): `"side-channel" AND agent`.

Table S21 provides the complete item-by-item reconciliation before and after the 204-record exclusion audit (177 high-risk pattern matches plus 27 reason-stratified samples, seed `20261021`). Before the exclusion audit, screening identified 221 C2/C3 exclusions (`EXCLUDE_NO_HIDDEN_AUXILIARY_TARGET`), 14 new core families, 23 existing core families, and 28 source-reviewed adjacent contexts. The exclusion audit rerouted exactly four records out of the 221 C2/C3 exclusions ($221 - 4 = 217$):

1) `SUP-0192` (MIRAGE [1]): rerouted to `NEW_CORE_SOURCE_VERIFIED` ($14 + 1 = 15$ final new core families, all 15 incorporated into the 82-family index via `addition.csv`).
2) `SUP-0552` (InputSnatch arXiv version [35]): rerouted to `EXISTING_CORE` ($23 + 1 = 24$ existing core recoveries; merged as an alias of the already-indexed snapshot family `inputsnatch2024`, adding $+0$ net new families).
3) `SUP-0191` (LLM Self-Recognition) and `SUP-0424` (Reverse-Engineering Model Editing): rerouted to `ADJACENT_SOURCE_REVIEWED` ($28 + 2 = 30$ final source-reviewed adjacent contexts). Of these 30 source-reviewed adjacent contexts, 7 boundary-defining studies (`addition-adjacent.csv`) are incorporated into the 82-family index ($17 + 7 = 24$ adjacent), while the other 23 (including `SUP-0191` and `SUP-0424`) remain in the 565-record disposition ledger as non-indexed context.

Thus the family transition is strictly additive and self-consistent: 60 (snapshot) + 15 (14 pre-audit + 1 audited new core) + 7 (selected adjacent) = 82 families ($43 + 15 = 58$ core; $17 + 7 = 24$ adjacent).

TABLE S20. REFERENCE-RELATIVE STATE EQUALITY ON 97 ELIGIBLE RETAIL TASKS. ENTRIES ARE PERCENTAGES WITH CONDITIONAL 95% TASK-BOOTSTRAP INTERVALS. NO-COVER, DISCARD, SHADOW, READ-ONLY AND ADDRESS-WRITE CONDITIONS ARE 100 [100,100] FOR ALL THREE SCOPES.

| Policy / position | Database equality | Full-account equality | Reference-changed-object equality |
|---|---|---|---|
| Item self-write before | 40.55 [31.62, 50.17] | 40.55 [31.62, 50.17] | 70.10 [61.51, 78.35] |
| Item self-write after | 67.70 [59.10, 76.29] | 67.70 [59.10, 76.29] | 97.25 [93.81, 99.66] |
| Instruction aware before | 35.05 [25.77, 44.33] | 35.05 [25.77, 44.33] | 78.35 [70.10, 86.60] |
| Instruction aware after | 46.39 [36.08, 56.70] | 46.39 [36.08, 56.70] | 89.69 [83.51, 94.85] |
| Other-account cancellation before | 0.00 [0.00, 0.00] | 100.00 [100.00, 100.00] | 100.00 [100.00, 100.00] |
| Other-account cancellation after | 0.00 [0.00, 0.00] | 100.00 [100.00, 100.00] | 100.00 [100.00, 100.00] |
| Blind cancellation before | 11.34 [5.84, 17.87] | 11.34 [5.84, 17.87] | 62.54 [53.61, 71.48] |
| Blind cancellation after | 41.24 [32.30, 50.17] | 41.24 [32.30, 50.17] | 92.44 [87.63, 96.56] |

TABLE S21. POST-AUDIT DISPOSITION FLOW FOR THE COMPLETED 565-RECORD SUPPLEMENTARY FRAME (SEE TABLE S3 FOR PRE-AUDIT VS. POST-AUDIT $\Delta$).

| Stage or final post-audit route | Count |
|---|---|
| Complete-query occurrences / duplicates merged | 615 / 50 |
| Unique records with an explicit route | 565 |
| Excluded: no hidden auxiliary target ($221 - 4$ rerouted) | 217 |
| Excluded: outside LLM execution / different target | 156 / 19 |
| Existing core ($23 + 1$ alias) / adjacent recovered | 24 / 1 |
| New source-bound core families ($14 + 1$ MIRAGE; all indexed) | 15 |
| Adjacent source-reviewed ($28 + 2$; 7 indexed) / metadata-only | 30 / 28 |
| Other security background / reviewed background / proposal | 53 / 4 / 1 |
| Missing abstract / unavailable primary full text | 16 / 1 |

## S10.2. Seed overlap clarification and dual-coding reliability

As noted in Main Paper Section 2, the 24 fixed known papers (`known24`) are a 24-paper core subset of our 40 initial citation-tracing seeds (`seed_n=24` in `check_known24_offline.py`), not an external holdout. Their 17/24 query recovery (and 25/60 snapshot recovery) measures lexical overlap between the six Boolean keyword queries and our seed-and-snowball corpus. Supplied second-coder annotations of a 100-record stratified subsample of the 565-record frame (`m7_dual_coding_screening.csv`) achieved 93.0% three-class agreement ($\kappa = 0.886$, 95% CI $[0.801, 0.963]$) and 95.0% binary Core-inclusion agreement ($\kappa = 0.880$, CI $[0.764, 0.975]$).

# S11. Retained buffered-TLS and fixed-position retail studies

This section preserves, unchanged in substance, the first-revision main-text account of the two controlled studies that Section 8 of the main paper now summarizes. Section S12 reports the second-round tests E5–E7.

The two studies test distinct observers. The transport observer sees TLS application-record lengths on a selected link. The tool-dispatch observer sees ordered tool names, write counts and response byte lengths, including failed calls; it is an instrumented dispatch view, not an encrypted-network observer. Neither experiment combines these views or runs an autonomous agent. Saved outputs and reference programs let us vary the specific boundary while holding the prescribed workload fixed.

## S11.1. Transport placement and workload-matched framing

For 24 public topics, the saved corpus contains one prompted brief and one prompted expanded `gpt-6-luna` response (48 outputs; 12 training and 12 disjoint test topics). These are designed prompt conditions, not a sample of natural task lengths. The original placement experiment passes text through two TLS 1.2 connections: the frontend decodes and removes upstream framing before sending new terminal chunks. Exact plaintext hashes are checked. Equalizing the upstream length projection leaves the unpadded terminal projection separable (100% verbosity classification); padding both projections reduces the evaluated classifier to 50%. This reproduces the receiving-boundary mechanism identified by NetEcho rather than discovering a new failure mode.

The revised framing replay uses `ECDHE-RSA-AES128-GCM-SHA256` sockets and parses application-data records. T1 contrasts brief versus full expanded responses. T2 contrasts the brief with the first two period-delimited sentences of the expanded response. T3 contrasts the first two versus first three sentences of the same expanded response. Thus T2/T3 are derived prefixes, with two examples per topic, not independently generated explanations. T4 relabels both brief and expanded responses by topic domain; it has 24 test outputs from 12 topics, not 12 expanded outputs. Its unpadded accuracy is 45.8% (95% topic-bootstrap interval 25.0–66.7%); its test majority-class baseline is 58.3%. T4 is a weak-signal control, not a defense success.

Every cost ratio divides defended application-record bytes by unpadded application-record bytes for the *same texts*, over all 48 examples in that workload. The count includes TLS record headers and encryption overhead; handshakes, TCP/IP, upstream bytes and latency are excluded. The 4-byte length prefix is included. Buckets of 256/512 bytes use 128-byte writes; 1024-byte buckets use 256-byte

TABLE S22. CORRECTED TLS APPLICATION-RECORD RESULTS. COSTS USE EACH WORKLOAD'S OWN UNPROTECTED TEXTS (ALL 48 OUTPUTS); ACCURACY USES 24 TEST OUTPUTS IN 12 TOPICS. INTERVALS RESAMPLE TOPICS.

**A. Two-link placement (T1; retained experiment)**

| Protection | Upstream acc. | Terminal acc. | Terminal bytes |
|---|---|---|---|
| None | 100.0 | 100.0 | 720.96 |
| Upstream only | 50.0 | 100.0 | 720.96 |
| Both, loose cap | 50.0 | 50.0 | 16,848 |

**B. Matched workload costs and defended-fit accuracy**

| Workload / framing | Mean bytes | Ratio | Acc. (%) |
|---|---|---|---|
| T1: unprotected | 720.96 | 1.00 | 100.0 |
| T1: 256B bucket | 856.96 | 1.19 | 100.0 |
| T1: 512B bucket | 1,085.92 | 1.51 | 100.0 |
| T1: 1024B bucket | 1,401.25 | 1.94 | 70.8 |
| T1: training-fitted cap | 1,570.00 | 2.18 | 50.0 |
| T1: loose cap | 16,848.00 | 23.37 | 50.0 |
| T2: unprotected | 272.48 | 1.00 | 100.0 |
| T2: 256B bucket | 438.29 | 1.61 | 87.5 |
| T2: 512B bucket | 628.00 | 2.30 | 50.0 |
| T2: 1024B bucket | 1,140.00 | 4.18 | 50.0 |
| T3: unprotected | 456.33 | 1.00 | 91.7 |
| T3: 256B bucket | 608.38 | 1.33 | 62.5 |
| T3: 512B bucket | 654.17 | 1.43 | 50.0 |
| T3: 1024B bucket | 1,140.00 | 2.50 | 50.0 |
| T4: unprotected | 720.96 | 1.00 | 45.8 |

**C. Adaptation on identical defended T1 tests**

| Defense | $D_0$ (%) | $D_2$ (%) | Gain [95% CI], pp |
|---|---|---|---|
| +512B offset | 50.0 | 100.0 | 50.0 [50.0, 50.0] |
| 1024B bucket | 50.0 | 70.8 | 20.8 [8.3, 33.3] |
| $U(0, 512)$B | 86.4 | 99.9 | 13.5 [10.7, 16.2] |
| $U(0, 1024)$B | 68.2 | 89.2 | 21.1 [17.2, 24.8] |
| Loose cap | 50.0 | 50.0 | 0.0 [0.0, 0.0] |
| Training-fitted cap | 50.0 | 50.0 | 0.0 [0.0, 0.0] |

Panel C uses 12 training pairs. Four-pair accuracy is the same except under $U(0, 1024)$B, where it is 90.2%. Random padding averages 50 draws within each topic; these are not 50 independent corpora. T2/T3 are sentence-prefix constructions. T4 relabels both original response lengths and has a 58.3% test majority baseline. The cap equalizes only the specified record-length projection when no message overflows.

writes. Consequently the 1024-byte single-quantum wire size is 1,140 bytes, not 1,082. The workload-matched 512-byte ratios for T2/T3 are 2.30×/1.43×; dividing both by T1's 720.96-byte denominator would yield the invalid earlier ratios below one.

A tighter comparison selects a constant application-frame cap using *training lengths only*: round the largest framed training response (1,194 bytes) up to a multiple of 128, yielding 1,280 bytes. Ten 128-byte records cost 1,570 wire bytes (2.18× T1). Longer messages would emit additional 128-byte records without truncation; no test message overflows. This produces identical length sequences on the saved T1 test set and 50% accuracy for the evaluated classifiers. It is a feasible cap for these data, not a globally optimal padding protocol or a guarantee for future lengths. The retained 16,848-byte cap (23.37×) is only a deliberately loose control.

## S11.2. Matched attacker adaptation

We use $D_0$ for an attacker fitted on unprotected training data and $D_2$ for one fitted on defended training data; both are evaluated on the identical defended test examples. The threshold uses total TLS record payload bytes. $D_2$ uses the first four or all twelve training topic pairs; these budgets do not represent multiple independently sampled training sets. At a 1024-byte bucket, $D_0$ constantly predicts the expanded class and scores 50%, while $D_2$ scores 70.8% (58.3–83.3%). An offset of 512 bytes permits 100% adapted accuracy. This demonstrates why a shifted $D_0$ alone is insufficient, without claiming novelty for adaptive evaluation.

Random-tail padding is evaluated over 50 draws per output. The first draw is socket measured and the remaining 49 use the record-length accounting checked against the socket replay. Each topic's paired correctness is averaged over its outputs and draws before 10,000 topic-bootstrap resamples (seed 20261021). Fractional accuracies therefore do not imply additional independent topics; the intervals condition on these saved outputs and fitted attackers. Positive adaptation gains span 13.5–50.0 percentage points; both constant caps have zero gain.

No measured accuracy of 50% proves absence of all leakage. The constant-cap construction makes only the specified record-length sequence constant when there is no overflow. Timing, connection behavior, request metadata and content are outside that projection. No sequence/timing attack, TLS 1.3 native padding, QUIC or live token/SSE stream is evaluated.

## S11.3. Retail execution: state scope and a visible-call counterexample

We replay the 115 public reference programs in the pinned $\tau$-bench retail simulator. There is no LLM planner or user simulator in this experiment. Ninety-seven tasks have an initially pending order owned by the task user; 18 have none and receive no cover. Three seeds select orders, and cover is inserted before or after the prescribed program. Utility uses 97 task clusters, each with three repeated units per position. Policies include no cover, discard, shadow execution, read only, address self-write, item self-write, instruction-aware selection, other-account cancellation and blind cancellation. Destructive conditions are local negative controls, not proposed deployment policies.

Separate the outcome scopes. Table S23 reports reference database equality, full active-account equality and equality on objects *changed* by the reference program. The last scope does not include every read/accessed object. The database criterion corresponds to the state part of the native evaluator; the extension script does not evaluate generated response correctness and we do not relabel its state equality as an end-to-end native task reward. All-account equality now checks every order owned by the active user and that user's record, rather than only initially pending orders.

Address self-writes preserve the reference state in this simulator. Item self-writes mutate state in 173/291 pre-

TABLE S23. RETAIL STATE AND OBSERVER RESULTS. STATE COLUMNS USE 97 ELIGIBLE TASKS (291 UNITS PER POSITION); OBSERVER COLUMNS USE 49 HELD-OUT TASKS (147 UNITS). PERCENTAGES ARE TASK MEANS ACROSS THREE SEEDS. B/A DENOTES BEFORE/AFTER.

**A. Distinct state scopes (reference execution as comparator)**

| Policy / position | DB eq. | Account eq. | Object eq. | Mutated |
|---|---|---|---|---|
| No cover B/A | 100.00 | 100.00 | 100.00 | 0.00 |
| Discard B/A | 100.00 | 100.00 | 100.00 | 0.00 |
| Shadow B/A | 100.00 | 100.00 | 100.00 | 0.00 |
| Read only B/A | 100.00 | 100.00 | 100.00 | 0.00 |
| Address self-write B/A | 100.00 | 100.00 | 100.00 | 0.00 |
| Item self-write B | 40.55 | 40.55 | 70.10 | 59.45 |
| Item self-write A | 67.70 | 67.70 | 97.25 | 32.30 |
| Instruction aware B | 35.05 | 35.05 | 78.35 | 70.10 |
| Instruction aware A | 46.39 | 46.39 | 89.69 | 53.61 |
| Other-account cancel B | 0.00 | 100.00 | 100.00 | 100.00 |
| Other-account cancel A | 0.00 | 100.00 | 100.00 | 100.00 |
| Blind cancel B | 11.34 | 11.34 | 62.54 | 100.00 |
| Blind cancel A | 41.24 | 41.24 | 92.44 | 58.76 |

**B. Legacy learned observer versus allowed-view rules**

| Policy / position | $D_0$ acc. | $D_2$ acc. | $D_2$ BA | Rule acc. |
|---|---|---|---|---|
| Majority (all conditions) | 77.55 | 77.55 | 50.00 | — |
| Instruction aware B | 51.02 | 73.47 | 47.37 | 100.00 |
| Instruction aware A | 51.02 | 71.43 | 52.51 | 100.00 |
| Other-account cancel B | 22.45 | 71.43 | 46.05 | 100.00 |
| Other-account cancel A | 22.45 | 71.43 | 46.05 | 100.00 |
| Blind cancel B | 22.45 | 71.43 | 46.05 | 100.00 |
| Blind cancel A | 22.45 | 68.03 | 47.09 | 100.00 |

DB eq.: equality of orders/users against reference (products are read-only), the native reward's database criterion, not an output-response/authorization check. Account eq.: all orders owned by the active user plus that user's record. Object eq.: objects changed by the reference program, not all accessed objects. Mutated: cover itself changes state. Other-account calls yield 0% DB equality but 100% active-account equality. BA: balanced accuracy. Rule: remove the known first/last cover call, then inspect cancellation verbs; blind/other-account cancellation also admits the $N_{\text{cancel}} \geq 2$ rule. Both rules attain 100% acc./BA in their stated conditions. Other non-cancellation policies retain 100% verb inference. Full intervals, AUROC and confusion matrices appear in Supplement S9 and the JSON artifact.

insertion units (59.45%): the tool changes the status to `pending (item modified)` and appends a refund entry even when the item identifier and price are unchanged. The remaining 118/291 calls fail because the requested variant is unavailable, explaining why this mutation rate is below blind cancellation's 291/291 successful calls. This is a state-preservation counterexample from one API, not a prevalence estimate or a discovery about idempotence. Formally, a no-change requirement $f(s, a) = s$ differs from repeated-call idempotence $f(f(s, a), a) = f(s, a)$.

Instruction-aware cover uses only instruction text and the initial order table for selection, with address self-write as fallback. It preserves reference-changed objects in 78.35%/89.69% of before/after units, but preserves the full account and database in only 35.05%/46.39%. Cancelling an unmentioned order can therefore damage a bystander resource. The heuristic was developed using the available task collection; it has no held-out policy-development evaluation. Its 68/97 availability (70.10%) and the item-self-write object equality of 204/291 (70.10%) are separate counts that happen to yield the same percentage.

The other-account condition selects a pending order belonging to another simulated user. It is not a provisioned dedicated decoy account. It preserves the active user's complete account and reference-changed objects in these tasks, while changing the shared database in every eligible unit (0% database equality). Thus account isolation of an effect cannot be described as whole-database preservation or an authorized production defense.

Observer permissions and simple attacks. The secret label is whether the reference program *attempts* a cancellation, including a call that may fail after cover execution. The observer sees the ordered dispatched tool names, write counts and response lengths; failed calls remain visible. It sees no arguments, user/order identifiers or database state. We do not infer that a passive TLS observer has these permissions. A backend with plaintext state would instead need a different confidentiality target.

The fixed task split contains 48 training tasks (7 positive, 41 negative) and 49 test tasks (11 positive, 38 negative); seeds stay within tasks. Always predicting no cancellation gives 77.55% test accuracy and 50% balanced accuracy. The legacy $D_2$ linear classifier uses cancellation presence, write count and write-response bytes. Its 68.03–73.47% accuracy under cancellation cover is below the majority baseline, and its balanced accuracy ranges from 46.05% to 52.51%. An increase over $D_0$ is therefore insufficient evidence for its claimed privacy interpretation.

For blind and other-account cancellation, exactly one extra cancellation is dispatched in every eligible session. The permitted rule $\widehat{y} = \mathbf{1}[N_{\text{cancel}} \geq 2]$ recovers the reference label at 100% accuracy and balanced accuracy on all 49 test tasks and three seeds, before and after insertion. For any policy with one live cover call at a known first/last position, removing that call and testing for a remaining cancellation also gives 100%. It needs no identifiers, no response content and no training labels. These are measured finite-sample results and a structural property of the fixed insertion design, not proof of general attack performance. Identifier filtering is unnecessary in this view and is not claimed as an evaluated attack.

Terminal equality does not establish path properties. Blind pre-insertion cancellation leaves the final database equal to reference in 33/291 units across 12 tasks, even though it preempts the later reference cancellation. The exact per-task counts are in Supplement S9. After insertion, 120/291 cover calls are rejected because reference execution already changed the order's status, and those rejections explain the 41.24% equality rate. Rejection is not successful benign cover. In the 82 tasks with no reference execution error (246 units), before/after equality is 24/246 (9.76%) and 94/246 (38.21%). These results illustrate the limits of endpoint scoring; consent and authorization were not measured.

## S12. Second-round transfer tests (E5–E7)

E5 and individual-attacker E6 results below retain the R2 measurements. Current E6/E7 suite summaries use post-review reanalysis in `experiments/sp2027-r3/`; historical registration and outputs remain in `experiments/sp2027-r2/`. A suite maximum is recomputed in each of 2,000 common task-bootstrap draws. Intervals condition on fixed out-of-fold predictions and omit retraining/search uncertainty. Meets: upper BA bound $< 0.60$ and maximum point AUROC $< 0.65$; Above: lower BA bound $> 0.60$; otherwise inconclusive. This is a finite-suite criterion, not a security guarantee.

TABLE S24. E5 PER ATTACKER WITH 95% TOPIC-BOOTSTRAP INTERVALS (12 TEST TOPICS). WIRE: MEAN APPLICATION-RECORD BYTES INCL. 5-BYTE HEADERS. AMB.: MEAN SIZE OF THE NEAREST-SEQUENCE TIE SET.

| TLS | Target | Framing | Wire | $D_0$ total | $D_2$ total | $D_2$ count | $D_2$ sequence | ID | Amb. |
|---|---|---|---|---|---|---|---|---|---|
| 1.2 | T1 | Buffered, unpadded | 721 | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 87.5 | 1.3 |
| 1.2 | T1 | Buffered, training cap | 1570 | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 2.1 | 48.0 |
| 1.2 | T1 | Stream, per token | 15038 | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 | 1.0 |
| 1.2 | T1 | + random event pad | 19194 | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 62.3 | 1.0 |
| 1.2 | T1 | + constant event size | 18255 | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 58.3 | 2.3 |
| 1.2 | T1 | 8-token batches, const. | 3040 | 50.0 [50.0,50.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 20.8 | 9.4 |
| 1.2 | T1 | Const. events + count cap | 36092 | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 6.3 | 44.1 |
| 1.2 | T3 | Buffered, unpadded | 456 | 91.7 [79.2,100.0] | 91.7 [79.2,100.0] | 91.7 [79.2,100.0] | 91.7 [79.2,100.0] | 91.7 | 1.2 |
| 1.2 | T3 | Buffered, training cap | 635 | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 4.2 | 44.2 |
| 1.2 | T3 | Stream, per token | 9713 | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 100.0 | 1.0 |
| 1.2 | T3 | + random event pad | 12398 | 71.2 [58.3,83.8] | 87.9 [75.8,99.2] | 87.5 [75.0,100.0] | 87.9 [76.2,98.3] | 72.5 | 1.0 |
| 1.2 | T3 | + constant event size | 11803 | 79.2 [66.7,91.7] | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 70.8 | 1.8 |
| 1.2 | T3 | 8-token batches, const. | 2023 | 50.0 [50.0,50.0] | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 18.8 | 8.4 |
| 1.2 | T3 | Const. events + count cap | 15411 | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 12.5 | 38.6 |
| 1.3 | T1 | Buffered, unpadded | 686 | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 87.5 | 1.3 |
| 1.3 | T1 | Buffered, training cap | 1500 | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 2.1 | 48.0 |
| 1.3 | T1 | Stream, per token | 14294 | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 | 1.0 |
| 1.3 | T1 | + random event pad | 18450 | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 62.3 | 1.0 |
| 1.3 | T1 | + constant event size | 17511 | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 58.3 | 2.3 |
| 1.3 | T1 | 8-token batches, const. | 2938 | 50.0 [50.0,50.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 20.8 | 9.4 |
| 1.3 | T1 | Const. events + count cap | 34626 | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 6.3 | 44.1 |
| 1.3 | T3 | Buffered, unpadded | 433 | 91.7 [79.2,100.0] | 91.7 [79.2,100.0] | 91.7 [79.2,100.0] | 83.3 [70.8,95.8] | 91.7 | 1.2 |
| 1.3 | T3 | Buffered, training cap | 606 | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 4.2 | 44.2 |
| 1.3 | T3 | Stream, per token | 9230 | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 100.0 | 1.0 |
| 1.3 | T3 | + random event pad | 11915 | 68.8 [56.7,81.7] | 87.9 [75.8,99.2] | 87.5 [75.0,100.0] | 87.5 [75.4,98.3] | 72.5 | 1.0 |
| 1.3 | T3 | + constant event size | 11321 | 79.2 [66.7,91.7] | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 70.8 | 1.8 |
| 1.3 | T3 | 8-token batches, const. | 1953 | 50.0 [50.0,50.0] | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 87.5 [75.0,100.0] | 18.8 | 8.4 |
| 1.3 | T3 | Const. events + count cap | 14782 | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 50.0 [50.0,50.0] | 12.5 | 38.6 |

TABLE S25. E6 DISPATCH VIEW: BALANCED ACCURACY (%) WITH TASK-CLUSTER BOOTSTRAP INTERVALS (115 TASKS, 3 SEEDS PER TASK), BEST AUROC, ACTIVE-ACCOUNT EQUALITY AND COVER-CALL ERROR RATE.

| Target | Cover | $D_0$ GB | $D_2$ LR | $D_2$ GB | Count rule | AUROC | Acct. eq. | Cover err. |
|---|---|---|---|---|---|---|---|---|
| cancel | No cover | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 1.00 | 100.0 | 0.0 |
| cancel | Cancel, $p$=0.5 | 74.6 [71.6,77.4] | 84.5 [75.9,92.1] | 86.1 [78.4,93.9] | 83.3 [75.0,91.2] | 0.95 | 100.0 | 0.0 |
| cancel | Cancel, $p$=1 | 50.0 [50.0,50.0] | 97.1 [92.4,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 1.00 | 100.0 | 0.0 |
| cancel | Cancel, $p$=1, live execution | 50.0 [50.0,50.0] | 98.1 [95.4,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 1.00 | 100.0 | 0.0 |
| cancel | Cancel count $\rightarrow$ 2 | 50.0 [50.0,50.0] | 76.9 [64.8,88.0] | 69.4 [57.6,81.7] | 50.0 [50.0,50.0] | 0.85 | 100.0 | 2.8 |
| cancel | One decoy program | 91.9 [89.8,94.2] | 68.1 [60.2,76.7] | 66.0 [60.1,72.2] | 91.9 [89.8,94.2] | 0.92 | 100.0 | 3.0 |
| cancel | Two decoy programs | 83.5 [80.5,86.5] | 55.9 [48.4,63.6] | 57.0 [49.9,64.1] | 83.5 [80.5,86.5] | 0.86 | 100.0 | 4.3 |
| cancel | Shape equalization | 50.0 [50.0,50.0] | 44.8 [37.6,52.4] | 49.2 [44.0,54.8] | 50.0 [50.0,50.0] | 0.50 | 100.0 | 5.7 |
| return | No cover | 100.0 [100.0,100.0] | 97.8 [94.0,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 1.00 | 100.0 | 0.0 |
| return | Cancel, $p$=0.5 | 100.0 [100.0,100.0] | 98.0 [94.3,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 1.00 | 100.0 | 0.0 |
| return | Cancel, $p$=1 | 100.0 [100.0,100.0] | 97.5 [94.8,99.6] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 1.00 | 100.0 | 0.0 |
| return | Cancel, $p$=1, live execution | 100.0 [100.0,100.0] | 98.0 [94.3,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 1.00 | 100.0 | 0.0 |
| return | Cancel count $\rightarrow$ 2 | 100.0 [100.0,100.0] | 98.0 [95.4,100.0] | 100.0 [100.0,100.0] | 100.0 [100.0,100.0] | 1.00 | 100.0 | 2.8 |
| return | One decoy program | 85.9 [82.9,89.0] | 73.2 [66.6,79.5] | 71.1 [65.5,76.8] | 85.9 [82.9,89.0] | 0.88 | 100.0 | 3.0 |
| return | Two decoy programs | 78.1 [75.0,81.2] | 70.6 [64.1,77.3] | 69.1 [62.9,75.0] | 78.1 [75.0,81.2] | 0.85 | 100.0 | 4.3 |
| return | Shape equalization | 50.0 [50.0,50.0] | 54.5 [48.4,60.3] | 50.3 [45.4,55.5] | 50.0 [50.0,50.0] | 0.53 | 100.0 | 5.7 |

TABLE S26. CORRECTED E6 SUITE MAXIMA AND TASK-BOOTSTRAP INTERVALS. INDIVIDUAL-ATTACKER RESULTS ABOVE ARE UNCHANGED. AUROC IS ALSO MAXIMIZED OVER THE EVALUATED SUITE.

| Target | Cover | Max BA [95% CI] | Max AUROC | Criterion |
|---|---|---|---|---|
| cancel | No cover | 100.0 [100.0,100.0] | 1.00 | Above |
| cancel | Cancel, $p$=0.5 | 86.1 [80.9,94.2] | 0.95 | Above |
| cancel | Cancel, $p$=1 | 100.0 [100.0,100.0] | 1.00 | Above |
| cancel | Cancel, $p$=1, live | 100.0 [100.0,100.0] | 1.00 | Above |
| cancel | Cancel count $\rightarrow$ 2 | 76.9 [67.0,88.1] | 0.85 | Above |
| cancel | One decoy | 91.9 [89.8,94.2] | 0.92 | Above |
| cancel | Two decoys | 83.5 [80.5,86.5] | 0.86 | Above |
| cancel | Shape | 50.0 [50.0,54.8] | 0.50 | Meets |
| return | No cover | 100.0 [100.0,100.0] | 1.00 | Above |
| return | Cancel, $p$=0.5 | 100.0 [100.0,100.0] | 1.00 | Above |
| return | Cancel, $p$=1 | 100.0 [100.0,100.0] | 1.00 | Above |
| return | Cancel, $p$=1, live | 100.0 [100.0,100.0] | 1.00 | Above |
| return | Cancel count $\rightarrow$ 2 | 100.0 [100.0,100.0] | 1.00 | Above |
| return | One decoy | 85.9 [82.9,89.0] | 0.88 | Above |
| return | Two decoys | 78.1 [75.0,81.2] | 0.85 | Above |
| return | Shape | 54.5 [50.0,60.4] | 0.53 | Inconcl. |

TABLE S27. CORRECTED E7 SUITE MAXIMA (4,140 SAVED TLS SESSIONS). DECODER: HELD-OUT REQUEST/RESPONSE SIZE-TO-TOOL ACCURACY; IT DIFFERS FROM REQUEST-ONLY IDENTIFICATION. COSTS USE RAW/NO-COVER AS DENOMINATOR. DEFENSE CALIBRATION USES THE FULL PUBLIC TASK POPULATION.

| Target | Framing | Cover | Max BA [95% CI] | AUROC | Criterion | Decoder | Bytes× | Calls× |
|---|---|---|---|---|---|---|---|---|
| cancel | raw | No cover | 100.0 [100.0,100.0] | 1.00 | Above | 92.8 | 1.00 | 1.00 |
| cancel | raw | Cancel count $\rightarrow$ 2 | 72.7 [63.4,84.5] | 0.81 | Above | 94.7 | 1.38 | 1.35 |
| cancel | raw | One decoy | 91.9 [89.8,94.2] | 0.92 | Above | 99.7 | 2.03 | 2.01 |
| cancel | raw | Shape | 51.2 [50.0,58.8] | 0.54 | Meets | 100.0 | 7.56 | 7.71 |
| cancel | bucket1024 | No cover | 60.1 [50.0,73.3] | 0.67 | Inconcl. | 35.2 | 2.42 | 1.00 |
| cancel | bucket1024 | Cancel count $\rightarrow$ 2 | 67.5 [58.5,78.9] | 0.70 | Inconcl. | 33.5 | 3.25 | 1.35 |
| cancel | bucket1024 | One decoy | 60.0 [53.4,68.7] | 0.59 | Inconcl. | 35.5 | 4.90 | 2.01 |
| cancel | bucket1024 | Shape | 55.3 [52.2,63.6] | 0.59 | Inconcl. | 26.5 | 18.56 | 7.71 |
| cancel | const | No cover | 59.7 [50.0,72.3] | 0.62 | Inconcl. | 30.4 | 3.41 | 1.00 |
| cancel | const | Cancel count $\rightarrow$ 2 | 64.9 [54.9,76.7] | 0.66 | Inconcl. | 30.0 | 4.61 | 1.35 |
| cancel | const | One decoy | 59.7 [52.6,67.4] | 0.59 | Inconcl. | 30.5 | 6.87 | 2.01 |
| cancel | const | Shape | 54.4 [50.0,62.5] | 0.50 | Inconcl. | 21.6 | 26.32 | 7.71 |
| return | raw | No cover | 95.7 [90.7,99.4] | 0.96 | Above | 93.8 | 1.00 | 1.00 |
| return | raw | Cancel count $\rightarrow$ 2 | 95.7 [90.6,99.4] | 0.96 | Above | 95.4 | 1.38 | 1.35 |
| return | raw | One decoy | 85.9 [82.9,89.0] | 0.88 | Above | 100.0 | 2.03 | 2.01 |
| return | raw | Shape | 60.5 [54.9,66.4] | 0.61 | Inconcl. | 100.0 | 7.56 | 7.71 |
| return | bucket1024 | No cover | 73.9 [66.6,83.1] | 0.76 | Above | 35.2 | 2.42 | 1.00 |
| return | bucket1024 | Cancel count $\rightarrow$ 2 | 66.0 [59.8,75.8] | 0.70 | Inconcl. | 33.5 | 3.25 | 1.35 |
| return | bucket1024 | One decoy | 59.1 [54.1,66.8] | 0.60 | Inconcl. | 35.5 | 4.90 | 2.01 |
| return | bucket1024 | Shape | 50.6 [50.0,57.3] | 0.52 | Meets | 26.5 | 18.56 | 7.71 |
| return | const | No cover | 53.6 [50.0,63.9] | 0.55 | Inconcl. | 30.4 | 3.41 | 1.00 |
| return | const | Cancel count $\rightarrow$ 2 | 59.1 [52.3,68.6] | 0.61 | Inconcl. | 30.0 | 4.61 | 1.35 |
| return | const | One decoy | 56.8 [51.4,63.6] | 0.57 | Inconcl. | 30.5 | 6.87 | 2.01 |
| return | const | Shape | 50.0 [50.0,55.1] | 0.50 | Meets | 21.6 | 26.32 | 7.71 |

TABLE S28. REQUEST-ONLY TOOL IDENTIFICATION: TRAINING-FOLD SIZE TABLE, UNSEEN-SIZE FALLBACK TO THE TRAINING MAJORITY. EACH OF 582 REFERENCE CALLS IS TESTED ONCE. THE 95.7% RAW IN-SAMPLE PURITY IS EXCLUDED FROM HELD-OUT ACCURACY.

| Task stratification | Framing | Held-out accuracy (%) | Training-majority baseline (%) |
|---|---|---|---|
| cancel | raw | 89.9 | 29.4 |
| cancel | bucket1024 | 29.4 | 29.4 |
| cancel | const | 29.4 | 29.4 |
| return | raw | 91.4 | 29.4 |
| return | bucket1024 | 29.4 | 29.4 |
| return | const | 29.4 | 29.4 |

Training-only tool-count maxima overflow on five distinct held-out tasks (22, 28, 40, 67, 87) under either target split. This is a feasibility audit, not a rerun of defense efficacy. Full fold membership and overflow records are released. No tested network configuration meets the criterion for both targets.

TABLE S29. CURRENT REASSESSMENT OF THE SEVEN AUTHOR-WRITTEN JUDGMENTS RECORDED BEFORE E5–E7. HISTORICAL OUTCOME LABELS ARE PRESERVED IN THE ARTIFACT, NOT COUNTED AS AUDIT ACCURACY.

| ID | Claim under examination | Current evidence and qualification |
|---|---|---|
| P1 | Buffered cost transfers to a stream | Emitter mismatch supported: per-event padding leaves T1 at 100%; constant-shape stream costs 2.42×. |
| P2 | Low $D_0$ establishes protection | Counterexample supported: batching and cancellation cover give $D_0$ 50%, $D_2$ 100%. |
| P3 | Per-event padding prevents identification | Counterexample supported: top-1 58–62% versus 2.1% chance in the closed candidate set. |
| P4 | Equal cancellation count hides cancellation | Residual leakage supported: count rule 50%; maximum BA 76.9%. |
| P5 | Dispatch protection transfers to the tool link | Protected-source premise unmet: count/decoy policies already leak at dispatch. Network BA 72.7–95.7% shows persistence. |
| P6 | Framing alone protects intent | Bucketed return remains above threshold; remaining framing-only cells inconclusive. |
| P7 | Joint cover/framing protects both targets | Not established: no configuration meets both target criteria. Joint constant framing: 50.0–54.4% BA; 26.3× bytes, 7.7× calls. |

## S13. Stand-in planner pipeline (E8) and online cover

`experiments/sp2027-r2/agent/` runs tool-calling agents and a user simulator on the 115 retail tasks, applies *online* cover (decided before each real call from the history only, with a final flush) and replays every session over TLS. Because sandboxed cover and framing never reach the agent, one trajectory per task and seed serves all policies. No LLM was available for this revision; the rows below use stand-in planners and exercise the pipeline. The scripted planner replays reference programs (native reward 1.00); the noisy planner inserts extra reads before each reference call with probability 0.3 (7.2 versus 5.1 real calls per session). These are synthetic results, not agent measurements.

TABLE S30. E8 DIAGNOSTICS REPORT POINT ESTIMATES ONLY; NO SUITE-MAXIMUM INTERVAL WAS RECOMPUTED FOR ONLINE RUNS. CURRENT OFFLINE E6/E7 INTERVALS ARE IN S12. BEST HELD-OUT BALANCED ACCURACY (%) POINT ESTIMATES FOR OFFLINE VERSUS ONLINE COVER. CONST.: CONSTANT FRAMING ON THE AGENT–TOOL LINK. CALLS+: COVER CALLS PER REAL CALL. OVERFLOW: SESSIONS WHOSE REAL CALLS EXCEED THE SHAPE TEMPLATE OR THE CANCELLATION TARGET.

| Planner / cover timing | Cover | Dispatch C | Dispatch R | Const. C | Const. R | Calls+ | Overflow (%) |
|---|---|---|---|---|---|---|---|
| Offline (E6/E7) | cancel const2 | 76.9 | 100.0 | 64.9 | 59.1 | 0.35 | 0.0 |
| Offline (E6/E7) | decoy1 | 91.9 | 85.9 | 59.7 | 56.8 | 1.01 | 0.0 |
| Offline (E6/E7) | shape eq | 50.0 | 54.5 | 54.4 | 50.0 | 6.71 | 0.0 |
| Online, scripted | cancel const2 | 79.6 | 100.0 | 63.0 | 59.1 | 0.36 | 3.8 |
| Online, scripted | decoy1 | 92.3 | 88.6 | 51.2 | 56.3 | 0.98 | 0.0 |
| Online, scripted | shape eq | 63.8 | 55.1 | 57.5 | 55.9 | 6.73 | 0.0 |
| Online, noisy scripted | cancel const2 | 79.1 | 100.0 | 66.0 | 60.0 | 0.25 | 4.1 |
| Online, noisy scripted | decoy1 | 92.3 | 88.6 | 51.8 | 56.2 | 0.69 | 0.0 |
| Online, noisy scripted | shape eq | 55.9 | 53.9 | 60.5 | 56.2 | 4.52 | 20.9 |

The preceding E8 rows and availability statement describe the historical stand-in study. S14 reports the subsequent real-model collection and training-only calibration.

## S14. Complete Real-Agent Evaluation with Training-Only Calibration (E9)

### S14.1. Cohort and execution

All 690 episodes in the frozen protocol are complete: 115 tasks, two prompt arms and three repetition identifiers. The fixed split assigns 69 tasks from 29 customers to training (414 episodes) and 46 tasks from 24 disjoint customers to testing (276 episodes). Every test task has all six prompt/repetition cells, and all failures and handoffs remain in the analysis. Training finished before defense calibration was frozen and test inference began. This section updates the earlier 663-episode publication snapshot to the full cohort; the immutable snapshot and the previous manuscript remain available in the archived release. No model configuration, prompt, split or defense parameter changed for this completion.

GPT-6.1 Sol with low reasoning effort supplies the planner and a separate user simulator through Codex app-server. The planner receives the retail policy, public tool schemas, interaction history and actual sandbox observations, and emits one structured action at each step. Reference actions and reward targets are used only for scoring. Each decision has a fresh ephemeral context without host tools or a host environment. The harness executes decisions in the pinned public retail simulator, with a 30-step limit. Repetition identifiers are cache namespaces; provider random seeds, temperature and an immutable weight snapshot are unavailable.

### S14.2. Prompts and defense calibration

The baseline prompt is the retail policy. The rules prompt adds one customer-details read after identification and before order access, an order read before each order write, and avoidance of unnecessary repeated reads. It preserves authorization and task requirements and prohibits unrelated operations for hiding intent. Both arms are evaluated under raw transport, padding only, online shape cover only, and joint shape cover with constant framing.

Only training trajectories from both prompts determine shared per-tool count maxima, donor pools and size caps. Donor calls exclude the active customer. Request caps are the largest training request plus four bytes rounded to a 256-byte multiple; response caps use the training 95th percentile plus four bytes rounded to a 1024-byte multiple, including training donor responses. Frozen caps are 1,024 and 3,072 bytes. The online dispatcher uses only past real-call counts, inserts a shadow call with probability 0.6 when a deficit exists and flushes remaining deficits at episode end. Overflow calls pass through; oversize messages expand to a 1024-byte bucket. Test data never recalibrate these parameters.

Shadow calls execute on isolated cloned state; their observations are suppressed and real calls/responses pass through unchanged. For deterministic tools with no timing-dependent side effects, projecting away cover gives the same planner-visible history under all program conditions. The replay therefore isolates transport intervention on the same realized actions, with reward preserved by the isolation invariant. All 690 real-call projections agree across the 2,760 saved sessions. Online means scheduling from past calls; interactive wall-clock delay and shared-resource feedback remain outside this replay measurement. A byte-transparent relay captures actual serialized requests and sandbox responses over local TLS 1.2.

### S14.3. Attackers and uncertainty

The frozen suite includes a training-majority baseline, clean-trained gradient boosting, defense-trained logistic regression and gradient boosting, a dispatch count rule, and network size-decoding learners and rules. All fitted parameters, thresholds and decoder tables use training data only. Targets are cancellation and return intent from the reference task; native task reward is scored separately using final database equality and required response strings. Points weight tasks equally. Intervals use 2,000 common customer-cluster bootstrap draws, retaining each customer's tasks and repetitions together and recomputing the suite maximum in each draw. Intervals condition on fitted attackers. Prompt contrasts pair task/repetition identifiers, without assuming identical conversations or shared model randomness.

### S14.4. Utility, overhead and prompt contrasts

Cancellation has 6 positive tasks from 5 customers; return has 11 positive tasks from 7 customers. Single-class draws excluded from 2,000 replicates: 6 for cancellation and 0 for return.

Baseline native reward is 110/138 (79.7%); rules reward is 105/138 (76.1%). All 24 baseline and 23 rules handoffs remain in the denominator. Joint-program cover-call error rates are 5.3% and 5.0%, respectively; all shape conditions have zero underfill. Overflow rates expose the resource demands of retaining real calls under the frozen training template.

TABLE S31. E9 DISPATCH VIEW: MAXIMUM BA (%) WITH 95% INTERVAL AND MAXIMUM AUROC ACROSS THE FIXED SUITE, ON 138 TEST EPISODES PER PROMPT. BA AND AUROC ARE SEPARATE SUITE MAXIMA.

| Prompt | Condition | Cancel BA [CI] | AUC | Return BA [CI] | AUC |
|---|---|---|---|---|---|
| baseline | Raw | 97.2 [90.0, 100.0] | 0.997 | 97.0 [88.9, 100.0] | 0.972 |
| baseline | Padding only | 97.2 [90.0, 100.0] | 0.997 | 97.0 [88.9, 100.0] | 0.972 |
| baseline | Shape only | 55.6 [50.0, 63.0] | 0.556 | 57.7 [50.0, 67.4] | 0.620 |
| baseline | Joint | 55.6 [50.0, 63.0] | 0.556 | 57.7 [50.0, 67.4] | 0.620 |
| rules | Raw | 97.2 [90.0, 100.0] | 0.972 | 97.0 [88.9, 100.0] | 0.970 |
| rules | Padding only | 97.2 [90.0, 100.0] | 0.972 | 97.0 [88.9, 100.0] | 0.970 |
| rules | Shape only | 56.5 [50.0, 68.3] | 0.655 | 64.7 [54.3, 77.4] | 0.693 |
| rules | Joint | 56.5 [50.0, 68.3] | 0.655 | 64.7 [54.3, 77.4] | 0.693 |

TABLE S32. E9 NETWORK VIEW: MAXIMUM BA (%) WITH 95% INTERVAL AND MAXIMUM AUROC ACROSS THE FIXED SUITE, ON 138 TEST EPISODES PER PROMPT. BA AND AUROC ARE SEPARATE SUITE MAXIMA.

| Prompt | Condition | Cancel BA [CI] | AUC | Return BA [CI] | AUC |
|---|---|---|---|---|---|
| baseline | Raw | 97.2 [90.0, 100.0] | 0.993 | 94.0 [85.8, 100.0] | 0.948 |
| baseline | Padding only | 66.1 [50.0, 83.9] | 0.634 | 55.5 [50.0, 73.0] | 0.567 |
| baseline | Shape only | 60.8 [55.4, 71.3] | 0.657 | 51.9 [50.5, 57.8] | 0.579 |
| baseline | Joint | 50.0 [50.0, 60.6] | 0.500 | 50.0 [50.0, 65.5] | 0.500 |
| rules | Raw | 97.2 [90.0, 100.0] | 0.978 | 93.5 [85.0, 99.5] | 0.937 |
| rules | Padding only | 67.6 [50.0, 85.0] | 0.636 | 57.9 [50.0, 78.9] | 0.548 |
| rules | Shape only | 52.8 [50.0, 65.5] | 0.588 | 53.2 [50.4, 63.9] | 0.544 |
| rules | Joint | 50.0 [50.0, 65.1] | 0.500 | 60.6 [50.0, 73.3] | 0.641 |

## S14.5. Interpretation and reproducibility

The completed cohort shows strong raw-trace intent inference and a large reduction under joint transport intervention. Rules improve mechanical adherence to the two read-order rules while leaving network BA above 93%. Baseline joint shaping reaches 50% BA for both targets, reducing the raw scores by 47.2 and 44.0 points. The finite-suite criterion requires the BA interval upper bound below 60% and point AUROC below 0.65. Baseline shape-only return meets it for that target; no condition meets it for both targets. Baseline joint upper bounds are 60.6%/65.5%, leaving the absolute-threshold conclusion inconclusive. The full-sample suite maximum equals the 50% majority baseline; several fitted attacks make heterogeneous predictions whose customer-resampled BA can exceed 50%. With only five cancellation-positive and seven return-positive customers, reselecting the maximum within each draw yields wider intervals. This mechanism does not imply a universal 10-point resolution floor or prove absence of residual population signal. The paired reduction intervals above exclude zero. Supplement S16 adds a task-permutation reference; the primary intervals remain clustered by customer. Rules-plus-joint return BA is 60.6%; its rules-minus-baseline contrast includes zero, as does the reward contrast.

The evidence covers one public retail environment and one mutable model alias, which also supplies user simulation. All 115 tasks were available during prior project development. The formal customer holdout therefore provides internal evaluation; external independent replication, broader model/environment transfer, provider-traffic measurement and interactive feedback effects remain follow-up questions.

The full cohort contains 11,586 model decisions, 2,760 TLS sessions and 176 attack-prediction sets. The frozen verifier passed after training-only recalibration, sandbox state replay, cover reconstruction, TLS accounting, metric/interval recomputation and request-provenance checks. Additional checks bind all agent actions and dialogue to actual model outputs and re-score required response strings. Offline attacker refitting reproduces the saved results and predictions exactly.

`experiments/sp2027-r4-full-20261023/` supplies full results, predictions and captures, plus the 27 new episodes and model records needed to extend the immutable 663-run snapshot. The three separately reported snapshot test episodes join the 27 new ones to expand the primary test from 246 to 276. `reproduce.py --reanalyze` reconstructs the full cohort from the base archives and delta, reruns analysis and executes the original frozen verifier without invoking a model. `verify_extension.py` checks decision alignment and provenance. The earlier snapshot, historical R2/R3 studies and pilot archives retain their original denominators and bytes.

TABLE S33. HELD-OUT UTILITY AND OVERHEAD. REWARD AND SESSION OVERFLOW RATES ARE PERCENTAGES; RATIOS USE RAW TRANSPORT WITHIN THE SAME PROMPT.

| Prompt | Condition | Reward [CI] | Calls× | Bytes× | Count overflow | Cap overflow | Underfill |
|---|---|---|---|---|---|---|---|
| baseline | Raw | 79.7 [69.0, 89.7] | 1.00 | 1.00 | 0.0 | 0.0 | 0.0 |
| baseline | Padding only | 79.7 [69.0, 89.7] | 1.00 | 4.02 | 0.0 | 6.5 | 0.0 |
| baseline | Shape only | 79.7 [69.0, 89.7] | 3.83 | 4.16 | 38.4 | 0.0 | 0.0 |
| baseline | Joint | 79.7 [69.0, 89.7] | 3.83 | 15.39 | 38.4 | 39.9 | 0.0 |
| rules | Raw | 76.1 [63.3, 88.7] | 1.00 | 1.00 | 0.0 | 0.0 | 0.0 |
| rules | Padding only | 76.1 [63.3, 88.7] | 1.00 | 4.04 | 0.0 | 6.5 | 0.0 |
| rules | Shape only | 76.1 [63.3, 88.7] | 3.88 | 4.27 | 44.2 | 0.0 | 0.0 |
| rules | Joint | 76.1 [63.3, 88.7] | 3.88 | 15.72 | 44.2 | 38.4 | 0.0 |

TABLE S34. MECHANICAL ADHERENCE TO THE TWO PROMPT-SPECIFIED READ-ORDER RULES (% WITH CUSTOMER-CLUSTER INTERVAL).

| Prompt | One successful customer read first | Order read before write | Both |
|---|---|---|---|
| baseline | 86.2 [74.5, 97.3] | 100.0 [100.0, 100.0] | 86.2 [74.5, 97.3] |
| rules | 97.1 [91.7, 100.0] | 100.0 [100.0, 100.0] | 97.1 [91.7, 100.0] |

TABLE S35. PAIRED PROMPT CONTRASTS USING COMMON CUSTOMER-BOOTSTRAP DRAWS.

| Rules minus baseline | Difference [95% CI], pp |
|---|---|
| Task reward | -3.6 [-9.5, 1.4] |
| Raw network, cancel BA | 0.0 [-8.3, 8.3] |
| Raw network, return BA | -0.5 [-2.6, 0.0] |
| Joint network, cancel BA | 0.0 [-1.7, 7.4] |
| Joint network, return BA | 10.6 [-13.3, 22.8] |

TABLE S36. POST HOC PAIRED PROGRAM CONTRASTS. THE FIXED ATTACK SUITE IS RE-MAXIMIZED UNDER EACH CONDITION WITHIN EACH COMMON CUSTOMER-BOOTSTRAP DRAW. ORIGINAL PREDICTIONS AND THE PREDECLARED ABSOLUTE CRITERION ARE UNCHANGED.

| Prompt | Target | Joint minus raw BA [95% CI], pp |
|---|---|---|
| baseline | cancel | -47.2 [-50.0, -33.3] |
| baseline | return | -44.0 [-48.9, -27.8] |
| rules | cancel | -47.2 [-50.0, -26.0] |
| rules | return | -33.0 [-44.3, -18.9] |

## S15. Cross-Fitting, Operating Points and Leakage Localization (E10–E12)

E10 retrospectively reuses the E9 defense design and attack suite under customer-grouped stratified 5-fold cross-fitting over all 690 completed episodes (115 tasks, 53 customers; 14 cancellation-positive and 18 return-positive customers). In each fold, the count template, donor pools, size caps and all attackers are fit only on the other folds. Out-of-fold predictions are pooled, and intervals use 2,000 customer-cluster bootstrap draws with the suite maximum recomputed per draw. The six analyzed configurations are specified in `r5lib.py`. E11 selects among them using inner customer folds of the 69 frozen training tasks only (rule in `e11_select.py`), then evaluates it on the 46 original test tasks. These analyses reuse an already analyzed cohort; only the original E9 protocol was frozen before test collection. E12 ablates undefended traces to locate the signal.

Records are computed with an analytic TLS 1.2 AES-GCM model (fragment plus 24 bytes, 16,384-byte fragments). The model is validated against all 2,760 socket captures of the completed cohort and 120 fresh loopback captures of new cross-fitted sessions. `verify_r5.py` also checks byte-identical reproduction of the E9 calibration, all sessions, and all 32 result cells of both the 660-run primary grid of the 663-run snapshot and the completed cohort. The same analysis on the 660-episode snapshot grid is retained in `snapshot-660/`; there the bucket point gives 50.6%/51.6% with upper bounds of 59.2%/58.3%. Cross-fitted templates use more training customers than E9 and are therefore larger, which raises call overhead (5.0× versus 3.8×). A quantile template is a natural next operating point.

TABLE S37. E10 CROSS-FITTED NETWORK VIEW: ALL CONFIGURATIONS AND PROMPTS. CORE = R4 SUITE; EXT. ADDS RANDOM FOREST, $k$-NN AND A FOREST SIZE DECODER. SA-AUC: SIGN-AGNOSTIC AUROC $\max(A, 1 - A)$.

| Configuration | Prompt | Cancel core [CI] | Ext. | SA-AUC | Return core [CI] | Ext. | SA-AUC | Bytes×/Calls× |
|---|---|---|---|---|---|---|---|---|
| R4 point (all tools, constant) (padding only) | baseline | 68.1 [55.4, 78.9] | 68.1 | 0.63 | 67.1 [57.2, 77.0] | 67.1 | 0.63 | 3.9/1.0 |
| R4 point (all tools, constant) | baseline | 60.5 [51.5, 70.5] | 60.5 | 0.59 | 52.1 [50.0, 60.3] | 52.7 | 0.63 | 19.8/5.0 |
| R4 point (all tools, constant) (padding only) | rules | 64.4 [54.6, 75.8] | 64.4 | 0.60 | 63.3 [52.7, 74.2] | 63.3 | 0.63 | 3.9/1.0 |
| R4 point (all tools, constant) | rules | 58.8 [50.0, 68.6] | 58.8 | 0.57 | 59.4 [50.5, 68.0] | 59.4 | 0.58 | 19.9/5.0 |
| All tools, 1024-B buckets | baseline | 52.2 [50.0, 60.5] | 52.2 | 0.54 | 52.7 [50.0, 60.2] | 52.7 | 0.61 | 12.4/5.0 |
| All tools, 1024-B buckets | rules | 53.4 [50.0, 60.6] | 53.4 | 0.57 | 50.7 [50.0, 57.3] | 52.4 | 0.56 | 12.4/5.0 |
| All tools, constant, q80 cap | baseline | 56.5 [50.5, 66.6] | 56.5 | 0.55 | 60.2 [55.0, 66.0] | 60.2 | 0.61 | 15.2/5.0 |
| All tools, constant, q80 cap | rules | 51.0 [50.0, 57.8] | 51.0 | 0.54 | 51.3 [50.0, 56.4] | 51.3 | 0.61 | 15.3/5.0 |
| Writes only, constant | baseline | 53.5 [50.0, 65.8] | 56.1 | 0.60 | 56.1 [50.0, 67.4] | 56.1 | 0.63 | 11.3/2.9 |
| Writes only, constant | rules | 62.6 [50.0, 74.4] | 62.6 | 0.60 | 57.6 [50.0, 68.2] | 57.6 | 0.58 | 11.4/2.9 |
| Writes only, constant, q80 cap | baseline | 53.7 [50.0, 66.4] | 53.7 | 0.55 | 62.0 [51.2, 73.3] | 62.0 | 0.66 | 8.6/2.9 |
| Writes only, constant, q80 cap | rules | 51.7 [50.0, 63.3] | 53.4 | 0.58 | 58.6 [50.0, 70.3] | 58.6 | 0.64 | 8.7/2.9 |
| Writes only, 1024-B buckets | baseline | 56.4 [50.0, 66.5] | 56.4 | 0.55 | 58.1 [50.1, 68.6] | 59.7 | 0.66 | 6.9/2.9 |
| Writes only, 1024-B buckets | rules | 52.6 [50.0, 63.8] | 52.6 | 0.60 | 58.7 [51.6, 69.6] | 58.7 | 0.59 | 7.0/2.9 |

TABLE S38. E11 INNER (TRAINING-FOLD) SELECTION STATISTICS. ELIGIBLE: ALL FOUR INNER NETWORK BAS $\leq 60\%$.

| Configuration | C/base | R/base | C/rules | R/rules | Bytes× | Eligible |
|---|---|---|---|---|---|---|
| R4 point (all tools, constant) | 54.9 | 53.5 | 58.4 | 52.6 | 15.2 | yes |
| All tools, 1024-B buckets | 53.3 | 58.1 | 55.3 | 56.8 | 9.8 | yes |
| All tools, constant, q80 cap | 56.8 | 56.8 | 53.1 | 54.8 | 11.7 | yes |
| Writes only, constant | 67.1 | 50.0 | 77.9 | 57.2 | 9.4 | no |
| Writes only, constant, q80 cap | 71.1 | 64.9 | 67.5 | 71.6 | 7.1 | no |
| Writes only, 1024-B buckets | 56.7 | 67.4 | 53.7 | 64.8 | 5.9 | no |

Rules-prompt homogenization: distinct read-only call sequences fall from 74 to 62 (entropy 5.63 to 5.37 bits), whereas state-changing sequences stay at 37 versus 36 (4.08 versus 4.09 bits).

TABLE S39. E12 OBSERVER ABLATION ON UNDEFENDED TRACES, CROSS-FITTED (CORE-SUITE BA, %).

| Observer | Prompt | View | Cancel | Return |
|---|---|---|---|---|
| full | baseline | dispatch | 91.7 | 93.8 |
| full | baseline | network | 91.7 | 85.5 |
| full | rules | dispatch | 90.7 | 93.8 |
| full | rules | network | 90.7 | 85.5 |
| no writes | baseline | dispatch | 53.1 | 65.7 |
| no writes | baseline | network | 55.2 | 59.9 |
| no writes | rules | dispatch | 60.8 | 66.2 |
| no writes | rules | network | 55.2 | 59.6 |
| writes only | baseline | dispatch | 91.7 | 93.8 |
| writes only | baseline | network | 91.7 | 84.9 |
| writes only | rules | dispatch | 90.7 | 93.8 |
| writes only | rules | network | 90.7 | 84.1 |

## S16. Conditional Task-Permutation Reference

This post hoc diagnostic holds the fitted E9 attackers fixed and permutes labels over the 46 test tasks 2,000 times (seed 20261023), keeping each task's three repetitions together. Each draw recomputes the attack-suite maximum. It characterizes a task-label exchangeability reference; it does not preserve dependencies among tasks belonging to one customer. Consequently, the tail fractions below are conditional reference values, not customer-valid population $p$-values or evidence of equivalence. The original customer-cluster intervals, paired contrasts and frozen absolute criterion remain the primary assessments.

TABLE S40. ALL 32 E9 CELLS: OBSERVED MAXIMUM BA, PERMUTATION MEAN AND 95TH PERCENTILE (%), CONDITIONAL TAIL FRACTION $(1 + \#\{T_b \geq T_{\mathrm{obs}}\})/2001$, AND POSITIVE EXCESS OVER THE 95TH PERCENTILE (PERCENTAGE POINTS).

| Prompt | View | Condition | Target | BA | Ref. mean | Ref. 95th | Tail | Excess |
|---|---|---|---|---|---|---|---|---|
| baseline | dispatch | Padding | cancel | 97.2 | 54.6 | 65.3 | 0.0005 | 31.9 |
| baseline | network | Padding | cancel | 66.1 | 55.2 | 70.8 | 0.0755 | 0.0 |
| baseline | dispatch | Padding | return | 97.0 | 54.1 | 65.1 | 0.0005 | 31.9 |
| baseline | network | Padding | return | 55.5 | 56.2 | 67.4 | 0.4883 | 0.0 |
| baseline | dispatch | Raw | cancel | 97.2 | 54.6 | 65.3 | 0.0005 | 31.9 |
| baseline | network | Raw | cancel | 97.2 | 55.5 | 68.5 | 0.0005 | 28.7 |
| baseline | dispatch | Raw | return | 97.0 | 54.1 | 65.1 | 0.0005 | 31.9 |
| baseline | network | Raw | return | 94.0 | 55.3 | 65.4 | 0.0005 | 28.7 |
| baseline | dispatch | Joint | cancel | 55.6 | 54.2 | 62.8 | 0.3048 | 0.0 |
| baseline | network | Joint | cancel | 50.0 | 54.7 | 66.2 | 1.0000 | 0.0 |
| baseline | dispatch | Joint | return | 57.7 | 53.0 | 58.7 | 0.1034 | 0.0 |
| baseline | network | Joint | return | 50.0 | 54.1 | 62.6 | 1.0000 | 0.0 |
| baseline | dispatch | Shape | cancel | 55.6 | 54.2 | 62.8 | 0.3048 | 0.0 |
| baseline | network | Shape | cancel | 60.8 | 54.9 | 63.3 | 0.0870 | 0.0 |
| baseline | dispatch | Shape | return | 57.7 | 53.0 | 58.7 | 0.1034 | 0.0 |
| baseline | network | Shape | return | 51.9 | 54.3 | 60.4 | 0.7956 | 0.0 |
| rules | dispatch | Padding | cancel | 97.2 | 54.6 | 68.5 | 0.0005 | 28.7 |
| rules | network | Padding | cancel | 67.6 | 55.7 | 69.2 | 0.1034 | 0.0 |
| rules | dispatch | Padding | return | 97.0 | 54.4 | 65.1 | 0.0005 | 31.9 |
| rules | network | Padding | return | 57.9 | 55.6 | 67.2 | 0.3033 | 0.0 |
| rules | dispatch | Raw | cancel | 97.2 | 54.6 | 68.5 | 0.0005 | 28.7 |
| rules | network | Raw | cancel | 97.2 | 55.1 | 68.5 | 0.0005 | 28.7 |
| rules | dispatch | Raw | return | 97.0 | 54.4 | 65.1 | 0.0005 | 31.9 |
| rules | network | Raw | return | 93.5 | 55.6 | 65.8 | 0.0005 | 27.8 |
| rules | dispatch | Joint | cancel | 56.5 | 53.2 | 59.7 | 0.1889 | 0.0 |
| rules | network | Joint | cancel | 50.0 | 53.8 | 63.2 | 1.0000 | 0.0 |
| rules | dispatch | Joint | return | 64.7 | 53.5 | 60.7 | 0.0115 | 4.0 |
| rules | network | Joint | return | 60.6 | 54.1 | 60.6 | 0.0680 | 0.0 |
| rules | dispatch | Shape | cancel | 56.5 | 53.2 | 59.7 | 0.1889 | 0.0 |
| rules | network | Shape | cancel | 52.8 | 54.3 | 60.8 | 0.6122 | 0.0 |
| rules | dispatch | Shape | return | 64.7 | 53.5 | 60.7 | 0.0115 | 4.0 |
| rules | network | Shape | return | 53.2 | 54.6 | 59.6 | 0.7081 | 0.0 |

The baseline raw network cells exceed the reference 95th percentiles (68.5%/65.4%), whereas baseline joint cells have no positive excess. This is consistent with the measured paired reductions; failure to exceed a reference threshold is not a population guarantee. The diagnostic uses saved predictions and makes its task-level exchangeability assumption explicit.

Complete indexed-family citation inventory. The following citation groups enumerate all 82 indexed families. Categories are core eligibility and origin, not mutually exclusive threat objectives. The machine-readable index binds each key to its source version and locators. Core (58 families): [31], [36], [37], [38], [32], [39], [40], [41], [35], [42], [43], [44], [45], [34], [46], [47], [48], [33], [49], [23], [2], [50], [51], [3], [4], [52], [5], [12], [53], [8], [9], [6], [27], [7], [10], [54], [11], [55], [13], [14], [56], [57], [58], [59], [60], [61], [62], [63], [1], [64], [65], [66], [67], [68], [69], [70], [71], [72].

Adjacent (24 families): [15], [73], [74], [16], [17], [18], [19], [20], [21], [22], [75], [76], [77], [28], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87].

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